A Project Report on

IOT BASED AGRICULTURE MONITORING SYSTEM

Submitted in partial fulfillment of the requirements of awards for the degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

BY

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

TIRUMALA ENGINEERING COLLEGE

Approved by AICTE, New Delhi & Affiliated to J.N.T.U. Kakinada

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Jonnalagadda (v), Narasaraopet, Palnadu (Dt)-522601

2020-2024

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CERTIFICATE

This is to certify that project work entitled **"IOT BASED AGRICULTURE MONITORING SYSTEM"** is the work carried out by **SK.B.HASEENA (20NE1A0219), A.SAI TEJA(21NE5A0220), M.YASHASWINI (20NE1A0211), S.TEJA(21NE5A0217), B.BALA BHASKAR NAIK(21NE5A0202)**, in fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ELECTRICAL AND ELECTRONICS ENGINEERING**, Certified by NBA, TIRUMALA ENGINEERING COLLEGE Narasaraopet is a bonafied project report carried by them under my guidance and supervision. The work has not been submitted to any other university of the award of any degree.

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DECLARATION

We are the students of Tirumala engineering college hereby declare that this project work entitled **"IOT BASED AGRICULTURE MONITORING SYSTEM"** being submitted by department of EEE, TMLN affiliated to **JNTU Kakinada**, for the award of bachelor of technology in **Electrical and Electronics Engineering** is a record of bonafied work done by us and it has not been submitted to any other institute our university for the award of any other degree or prize.

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ABSTRACT

Smart farming, precision agriculture and Agriculture all involve the integration of advanced technologies into existing farming architecture. The goal is to increase production efficiency and product quality, as well as reducing overall costs. To this end, the inclusion of Smart technologies into Irish agriculture has been inevitable with increased pressure beingplaced on farming practices to remain profitable, as well as adhere to environmental regulation. The global Smart Agriculture Solution Market is said to have stood at around US $10.2 Billionin 2016, and is projected to reach a valuation of US $38.1 Billion by the end of 2024.

The growing adoption of advanced technology in farming, from agricultural drones, precision seeding systems, auto-steering, automatic feeding systems and fruit-picking robots (amongst others), have all incentivised traditional agri-companies to invest in smart agriculture technology.

The deployment of advanced agri-tech has the potential to allow for an increased focus on non-profitable tasks, such as farm maintenance and environmental practices. The reduction of heavy labour and tedious tasks can also lead to improvements in the health and work/life balance of farming staff.

Moreover, the advent of agricultural drones offers unprecedented capabilities for crop monitoring, pest detection, and even aerial spraying, enhancing efficiency and accuracy in farm management. These drones can cover vast areas in a fraction of the time it would take traditional methods, allowing farmers to detect issues early and take proactive measures to mitigate potential losses.

Automatic feeding systems and fruit-picking robots revolutionize labour-intensive tasks, relieving farmers from repetitive and physically demanding work. By automating these processes, farmers can allocate their time and resources more efficiently, focusing on tasks that require human expertise and creativity.

Overall, the adoption of smart agriculture technologies not only enhances productivity and profitability but also promotes sustainability, resilience, and improved quality of life for farming communities. As the global population continues to grow, and environmental challenges become more pressing, the role of advanced technologies in agriculture will be increasingly vital in ensuring food security and environmental sustainability for future generations.

CHAPTER-1

INTRODUCTION

1.1 : General Background

With the exponential growth of world population, according to the UN Food and Agriculture Organization, the world will need to produce 70% more food in 2050, shrinking agricultural lands, and depletion of finite natural resources, the need to enhance farm yield has become critical. Limited availability of natural resources such as fresh water and arable land along with slowing yield trends in several staple crops, have further aggravated the problem. Another impeding concern over the farming industry is the shifting structure of agricultural workforce. Moreover, agricultural labour in most of the countries has declined.

As a result of the declining agricultural workforce, adoption of internet connectivity solutions in farming practices has been triggered, to reduce the need for manual labour. IoT solutions are focused on helping farmers close the supply demand gap, by ensuring high yields, profitability, and protection of the environment. The approach of using IoT technology to ensure optimum application of resources to achieve high crop yields and reduce operational costs is called precision agriculture.

IoT in agriculture technologies comprise specialized equipment, wireless connectivity, software and IT services. BI Intelligence survey expects that the adoption of IoT devices in the agriculture industry will reach 75 million in 2020, growing 20% annually. At the same time, the global smart agriculture market size is expected to triple by 2025, reaching $15.3 billion (compared to being slightly over $5 billion back in 2016).

Smart farming based on IoT technologies enables growers and farmers to reduce waste and enhance productivity ranging from the quantity of fertilizer utilized to the number ofjourneys the farm vehicles have made, and enabling efficient utilization of resources such as water, electricity, etc. IoT smart farming solutions is a system that is built for monitoring the crop field with the help of sensors (light, humidity, temperature, soil moisture, crop health, etc.) and automating the irrigation system. The farmers can monitor the field conditions from anywhere. They can also select between manual and automated options for taking necessary actions based on this data. For example, if the soil moisture level decreases, the farmer can deploy sensors to start the irrigation. Smart farming is highly efficient when compared with the conventional approach. IoT have the potential to transform agriculture in many aspects and these are the main ones. Data collected by smart agriculture sensors, in this approach of farm management, a key component are sensors, control systems, robotics, autonomous vehicles, automated hardware, variable rate technology, motion detectors, button camera, and wearable devices. This data can be used to track the state of the business in general as well as staff performance, equipment efficiency.

The ability to foresee the output of production allows to plan for better product distribution. Agricultural Drones Ground-based and aerial-based drones are being used in agriculture in order to enhance various agricultural practices: crop health assessment, irrigation, crop monitoring, crop spraying, planting, and soil and field analysis.

1.2 : Related works

An IOT Based Crop-field monitoring an irrigation automation system describes how to monitor a crop field. A system is developed by using sensors and according to the decision from a server based on sensed data, the irrigation system is automated. Through wireless transmission the sensed data is forwarded to web server database. If the irrigation is automated then the moisture and temperature fields are decreased below the potential range. The user can monitor and controlthe system remotely with the help of application which provides a web interface to user.

By smart Agriculture monitoring system and one of the oldest ways in agriculture is the manual method of checking the parameters. In this method farmers by themselves verify all the parameter and calculate the reading.

The system focuses on developing devices and tool to manage, display and alert the users using the advantages of a wireless sensor network system. It aims at making agriculture smart using automation and IoT technologies.

The cloud computing devices are used at the end of the system that can create a whole computing system from sensors to tools that observe data from agriculture field. It proposes a novel methodology for smart farming by including a smart sensing system and smart irrigator system through wireless communication technology. This system is cheap at cost for installation.Here one can access and also control the agriculture system in laptop, cell phone or a computer.

IoT-based smart agriculture is a modern approach to farming that leverages Internet of Things (IoT) technology to enhance agricultural practices. Here's a breakdown of the existing system and the proposed system for IoT-based smart agriculture.

1.3: Existing System

1. Traditional Agriculture: Before the implementation of IoT, agriculture relied on traditional methods and practices, which often led to inefficiencies and increased resource consumption.

2. Limited Data: Farmers relied on their experience and knowledge to make decisions about planting, irrigation, and pest control. Data collection was minimal, making it challenging to optimize crop yields and resource usage.

3. Manual Monitoring: Monitoring of environmental factors such as soil moisture, temperature, and humidity was primarily done manually, which was time-consuming and often prone to errors.

4. Resource Inefficiencies: Water and energy resources were often wasted due to lack of precise monitoring and control systems.

1.4: Summary

When the IOT based agriculture monitoring system starts it checks the water level, humidity and moisture level. It sends SMS alert on the phone about the levels. Sensors sense the level of water if it goes down, it automatically starts the water pump. If the temperature goes above the level, fan starts. This system also minimizes human efforts, simplifies techniques of farming and helps to gain smart farming.

CHAPTER-2

IMPLEMENTATION OF PROJECT

2.1: PROPOSED SYSTEM

2.1.1: Proposed System (IoT-based Smart Agriculture):

1. IoT Sensors: Implementing a network of IoT sensors and devices throughout the farm to collect real-time data on various parameters like soil moisture, temperature, humidity, and crop health.

2. Data Analytics: Utilizing data analytics and machine learning algorithms to process the data collected from sensors. This helps in making informed decisions regarding irrigation, fertilization, and pest control.

3. Remote Monitoring: Farmers can remotely monitor their crops and farm conditions through smartphones or computers, allowing them to react quickly to changes in environmental conditions.

4. Resource Conservation: IoT-based smart agriculture aims to optimize resource usage, conserve water, reduce energy consumption, and minimize the environmental impact of farming.

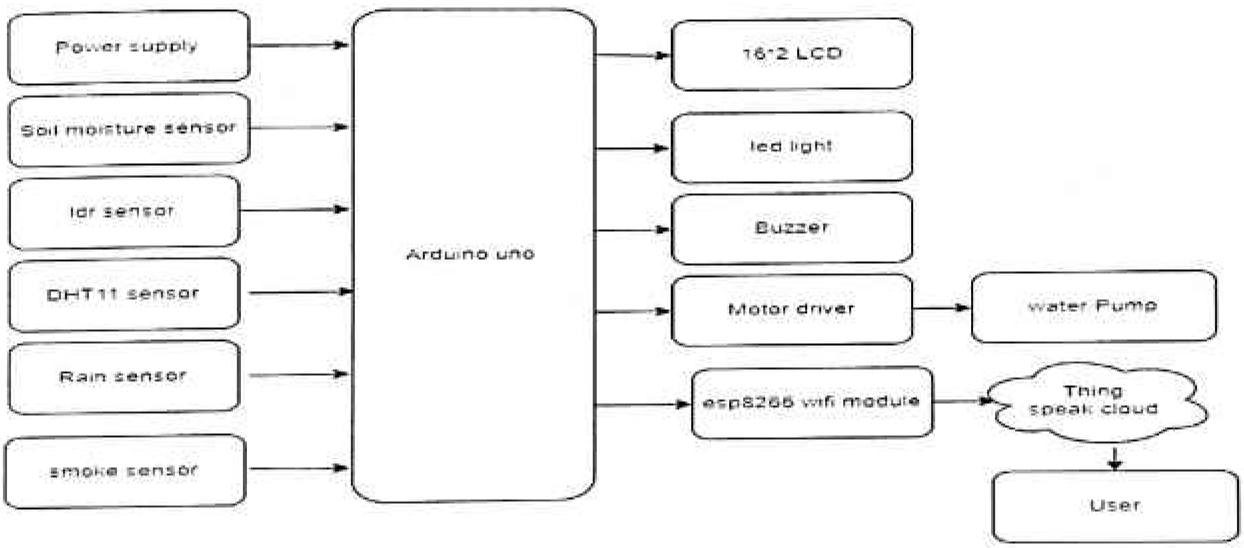


Fig 2.1.1 Block Diagram IoT based Agriculture Monitoring System

2.2: Description

The smart agriculture monitoring system is tested under various conditions. The soil moisture sensor is used to test the soil for all climatic conditions and results are interpreted successfully. The moisture output readings at different weather conditions is taken and updated. Wi-Fi is used to achieve the wireless transmission. The values of soil moisture sensor purely depend on the resistivity of the soil. The value of the sensor at beginning of wet condition is 0. The sensed value is sent to microcontroller through Node MCU and motor pump gets OFF in this condition. The maximum threshold value upon dry soil is 1023. When the sensed value by sensor reaches the threshold value, the microcontroller trigger the relay and motor gets ON. When sufficient amount of water is supplied to plants, the motor pump is turned ON and is turned OFF automatically.

2.3: Advantages and Applications

It is easy to maintain and cost is reasonable to purchase. The components which are used are easily available.

- It has advantage to observe the status on smartphone or laptop using internet. The information is up to date even in absence of farmer.
- The collected data is updated and the farmer is conscious about the status of the crop.
- To achieve more effective and accurate details of crop several additional sensors can also be included.
- Precision agriculture is one of the most popular application of IoT in the agricultural sector.
- High-speed internet, mobile devices, and affordable satellites are key technologies driving the precision agriculture trend among manufacturers.
- Precision agriculture or precision farming, is any method that enhances the control and accuracy of raising livestock and growing crops.
- IoT-based smart farming applications enhance not only conventional, large-scale operations but also support trends like organic, family, and transparent farming.

2.4: Summary

The system uses various sensors to monitor environmental conditions in real-time. The data collected is processed by a microcontroller and transmitted wirelessly to a web application that provides farmers with visualized information about their crops.

CHAPTER-3

LITERATURE REVIEW OF COMPONENTS

3.1: HARDWARE DESCRIPTION

3.1.1 : Arduino

Arduino is open-source physical processing which is based on a microcontroller board and an incorporated development environment for the board to be programmed. Arduino gains a few inputs, for example, switches or sensors and control a few multiple outputs, for example, lights, engine and others. Arduino program can run on Windows, Macintosh and Linux operating systems (OS) opposite to most microcontrollers' frameworks which run only on Windows. Arduino programming is easy to learn and apply to beginners and amateurs. Arduino is an instrument used to build a better version of a computer which can control, interact and sense more than a normal desktop computer. It's an open-source physical processing stagefocused around a straightforward microcontroller board, and an environment for composing programs for the board. Arduino can be utilized to create interactive items, taking inputs from a diverse collection of switches or sensors, and controlling an assortment of lights, engines, and other physical outputs. Arduino activities can be remaining solitary, or they can be associated with programs running on your machine (e.g. Flash, Processing and Maxmsp.) The board can be amassed by hand or bought preassembled; the open-source IDE can be downloaded free of charge. Focused around the Processing media programming environment, the Arduino programming language is an execution of Wiring, a comparative physical computing platform.

Figure 3.1.1 Arduino Symbol

3.1.2 : Why choosing Arduino

There are numerous different microcontrollers and microcontroller platforms accessible for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and numerous others offer comparative usefulness.

These apparatuses take the chaotic subtle elements of microcontroller programming and wrap it up in a simple to-utilize bundle. Arduino additionally rearranges the methodology of working with microcontrollers; moreover, it offers some advantages for instructors, students, and intrigued individuals:

- Inexpensive - Arduino boards are moderately cheap compared with other microcontroller boards. The cheapest version of the Arduino module can be amassed by hand, and even the preassembled Arduino modules cost short of what $50.
- Cross-platform - The Arduino programming runs multiple operating systems Windows, Macintosh OSX, and Linux working frameworks. So we conclude that Arduino has an advantage as most microcontroller frameworks are constrained to Windows.
- Straightforward, clear programming method - The Arduino programming environment is easy to use for novices, yet sufficiently versatile for cutting edge customers to adventure as well. For educators, its favourably engaged around the Processing programming environment, so under studies finding ways to understand how to program in that environment will be familiar with the nature of Arduino.
- Open source and extensible programming-The Arduino program language is available as open source, available for development by experienced engineers. The lingo can be reached out through C++ libraries, and people expecting to understand the specific purposes of different interests can make the leap from Arduino to the AVR C programming language on which it is based. Basically, you can incorporate AVR-C code clearly into your Arduino programs if you have to.
- Open source and extensible hardware - The Arduino is concentrated around Atmel's Atmega8 and Atmega168 microcontrollers.
- The plans for the modules are circulated under a Creative Commons license, so experienced circuit designers can make their own particular interpretation of the module, extending it and improving it. slightly inexperienced customers can build the breadboard variation ofthe module remembering the finished objective to perceive how it capacities and save money.

3.2 : Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328(datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer witha USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega8U2 programmed as a USB-to-serial converter. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduinoboards, and the reference model for the Arduino platform.

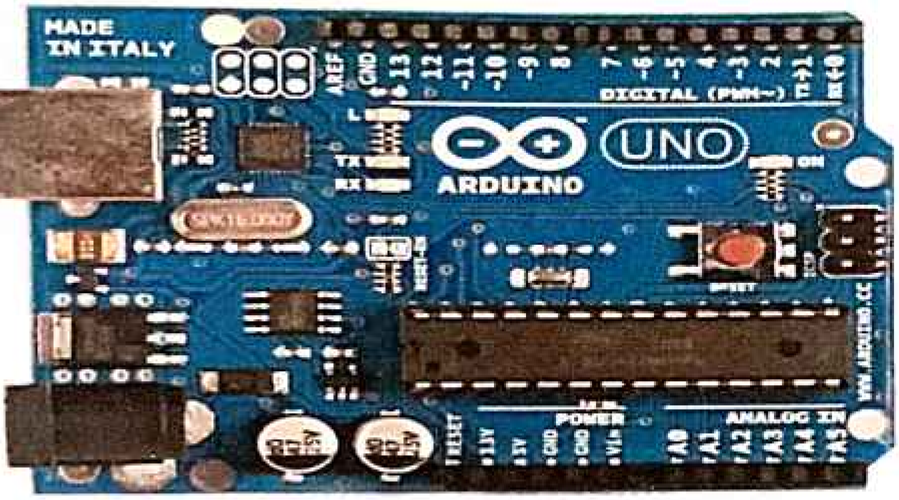


Fig 3.2. Arduino Uno

3.2.1 : Technical specifications of Arduino:

- Microcontroller: ATmega328
- Operating Voltage: 5V
- Input Voltage (recommended): 7-12V
- Input Voltage (limits): 6-20V
- Digital I/O Pins 14 (of which 6 provide PWM output)
- Analog Input Pins 6
- DC Current per I/O Pin 40 mA
- DC Current for 3.3V Pin 50 mA
- Flash Memory

- 32 KB of which 0.5 KB used by
- bootloader
- SRAM 2 KB
- EEPROM 1 KB
- Clock Speed 16 MHz

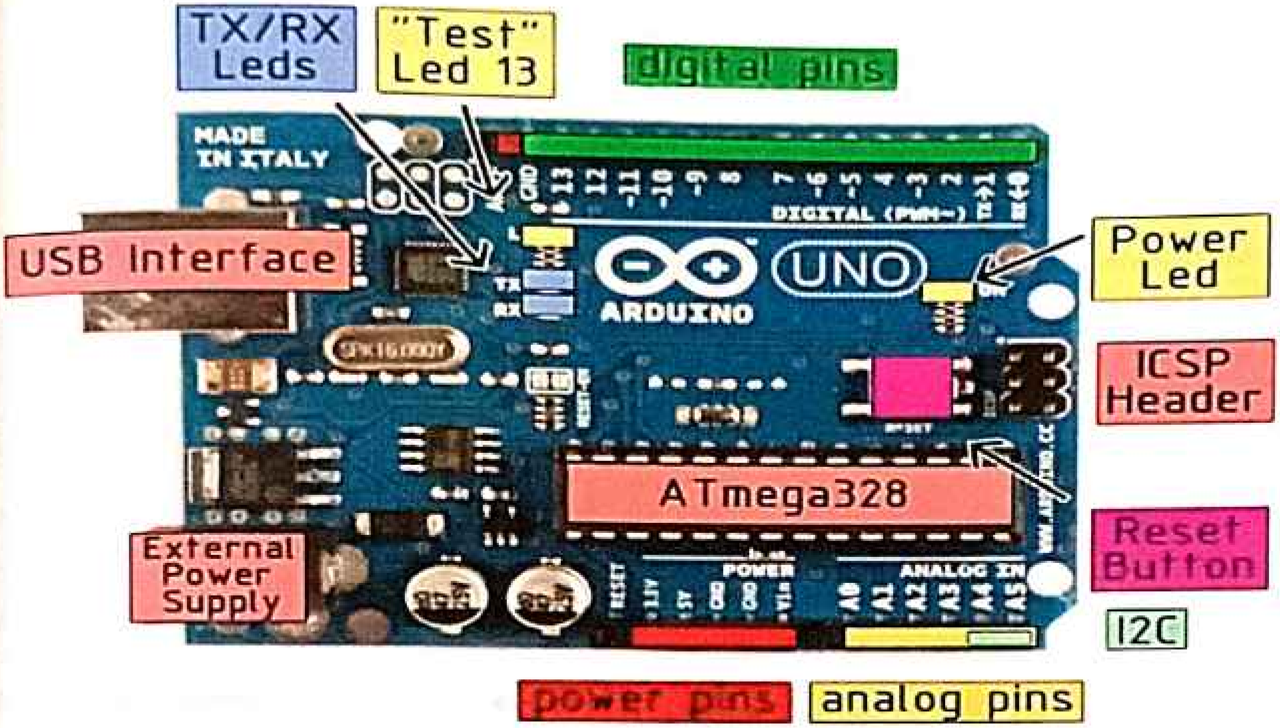


Fig 3.2.1: Parts of Arduino Uno

3.2.2 : Power

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- Vin: The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it throughthis pin.
- 5V: The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- 3V3: A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND:** Ground pins.

3.2.3 : Memory

The Atmega328 has 32 KB of flash memory for storing code (of which 0,5 KB is used for the bootloader); It has also 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

3.2.4 : Input/Output

Each of the 14 digital pins on the Uno can be used as an input or output, using pinMode(), digital Write(), and digital Read() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kiloohms. In addition, some pins have specialized functions:

- Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip,
- External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach Interrupt() function for details.
- PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analog Write() function.

- SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.
- LED: 13. There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off. The Uno has 6 analog inputs, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from groundto 5 volts, though is it possible to change the upper end of their range using the AREF pin and the analog Reference() function. Additionally, some pins have specialized functionality:
- I2C: 4 (SDA) and 5 (SCL), Support I2C (TWI) communication using the Wire library. There are a couple of other pins on the board:
- AREF: Reference voltage for the analog inputs. Used with analog Reference().
- Reset: Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

3.3 : Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega8U2 on the board channels this serial communication over USB and appears as a virtual com port tosoftware on the computer. The '8U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, an *.inf file is required.

The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1). A Software serial library allows for serial communicationon any of the Uno's digital pins.

The ATmega328 also support I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus see the documentation for details. To use the SPI communication, please see the ATmega328 datasheet.

3.4 : Programming

The Arduino Uno can be programmed with the Arduino software. The ATmega328 on the Arduino Uno comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files). You can also bypass the bootloader and program the microcontroller through the ICSP (In Circuit Serial Programming) header; see these instructions for details. The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available. The ATmega16U2/8U2 is loaded with a DFU bootloader, which can beactivated by: On Rev1 boards: connecting the solder jumper on the back ofthe board (near the map of Italy)and then resetting the 8U2. 10 On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2HWB line to ground, making it easier to put into DFU mode. You can then use Atmel's FLIP software (Windows) or the DFU programmer (Mac OS X and Linux) to load a new firmware. Or you can use the ISP header with an external programmer (overwriting the DFU bootloader). See this user-contributed tutorial for more information.

3.5 : Automatic Software Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino Uno is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of theATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino software uses this capability to allow you to upload code by simply pressing the upload button in the Arduino environment. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well- coordinated with the start ofthe upload. This setup has other implications. When the Uno is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it fromsoftware (via USB).

For the following half-second or so, the bootloader is running on the Uno. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data. The Uno contains a trace that can be cut to disable the auto-reset. The pads on either side ofthe trace can be soldered together to re-enable it. It's labelled "RESET- EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V tothe reset line 11.

3.6 : USB Overcurrent Protection

The Arduino Uno has a resettable poly fuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed. Additionally, the resettable poly fuse on the Arduino Uno serves as a safeguard against potential damage to both the board and connected devices by interrupting the power flow in the event of excessive current draw.

3.7 : Physical Characteristics

The maximum length and width of the Uno PCB are 2.7 and 2.1 inches respectively, with the USB connector and power jack extending beyond the former dimension. Four screw holes allow the board to be attached to a surface or case. Note that the distance between digital pins 7 and 8 is 160 mil (0.16"), not an even multiple of the 100 mil spacing of the other

3.8 : Power Supply

Power supply is a reference to a source of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

This power supply section is required to convert AC signal to DC signal and also to reduce the amplitude of the signal. The available voltage signal from the mains is 230V/50Hz which is an AC voltage, but the required is DC voltage (no frequency) with the amplitude of
+5V and +12V for various applications.

In this section we have Transformer, Bridge rectifier, are connected serially and voltage regulators for +5V and +12V (7805 and 7812) via a capacitor (1000µF) in parallel are connected parallel as shown in the circuit diagram below. Each voltage regulator output is again is connected to the capacitors of values (100µF, 10µF, 1 µF, 0.1 µF) are connected parallel throughwhich the corresponding output (+5V or +12V) are taken into consideration.

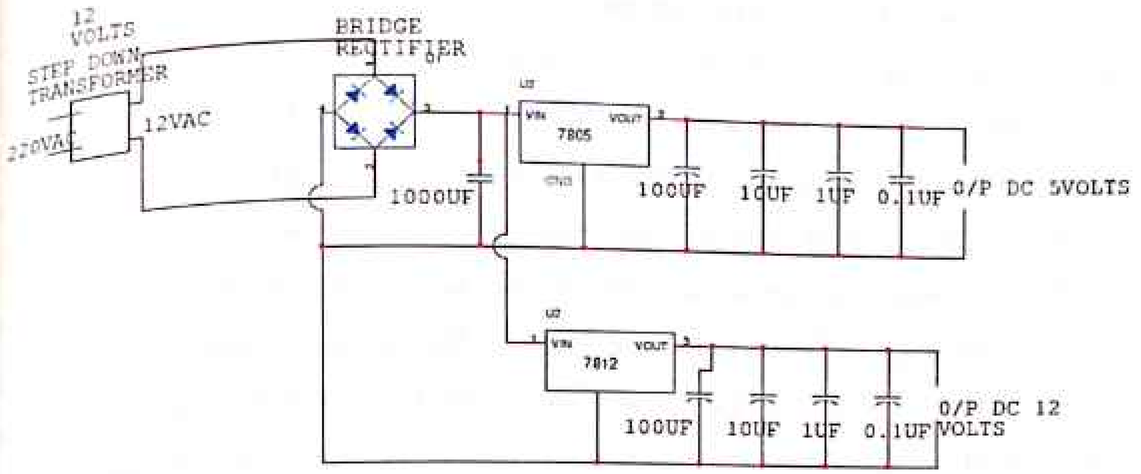


Fig 3.8. Circuit Diagram of Bridge Rectifier

3.8.1 : Circuit Explanation

3.8.1.1 : Transformer

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled electrical conductors. A changing current in the first circuit (the primary) creates a changing magnetic field; in turn, this magnetic field induces a changing voltage in the second circuit (the secondary). By adding a load to the secondary circuit, one can make current flow in the transformer, thus transferring energy from one circuit to the other.

The secondary induced voltage V_S, of an ideal transformer, is scaled from the primary V_P by a factor equal to the ratio of the number of turns of wire in their respective windings:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

3.8.1.2 : Basic principle

The transformer is based on two principles: firstly, that an electric current can produce a magnetic field (electromagnetism) and secondly that a changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). By changing the current in the primary coil, it changes the strength of its magnetic field since the changing magnetic field extends into the secondary coil, a voltage is induced across the secondary.

A simplified transformer design is shown below. A current passing through the primary coil creates a magnetic field. The primary and secondary coils are wrapped around a core of very high magnetic permeability, such as iron; this ensures that most of the magnetic field lines produced by the primary current are within the iron and pass through the secondary coil as well as the primary coil. The core's high permeability helps concentrate the magnetic flux, enhancing the transformer's efficiency by minimizing losses due to magnetic leakage. The ratio of the number of turns in the primary coil to the number of turns in the secondary coil determines the voltage transformation ratio. If the secondary coil has more turns than the primary, the transformer steps up the voltage, and if it has fewer turns, the voltage is stepped down. This principle enables transformers to adapt voltage levels to suit various applications, from power distribution to electronics, making them indispensable in modern electrical systems. Additionally, transformers also provide isolation between the primary and secondary circuits, which is crucial for safety and noise reduction.

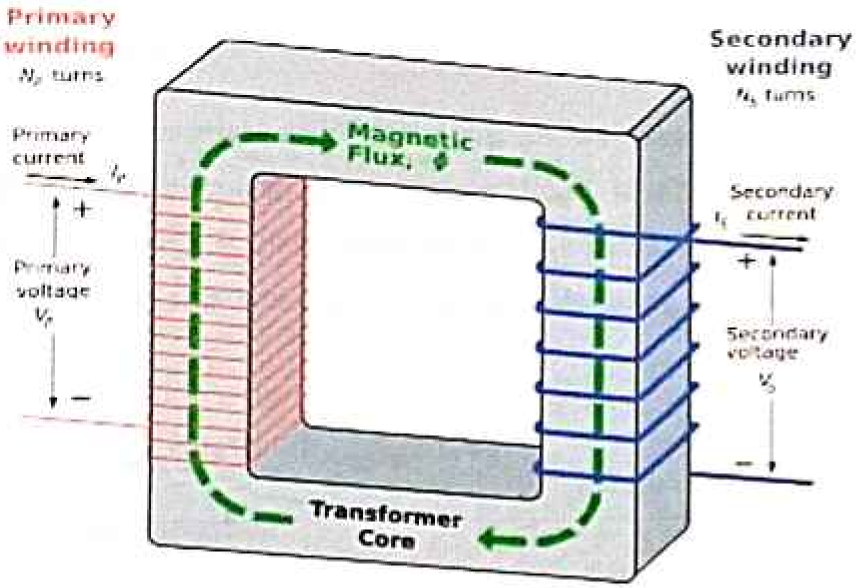


Fig 3.8.1.2: An ideal step-down transformer showing magnetic flux in the core

3.8.1.3 : Induction law

The voltage induced across the secondary coil may be calculated from Faraday's law of induction, which states that:

$$V_S = N_S \frac{d\Phi}{dt}$$

Where Vs is the instantaneous voltage, Ns is the number of turns in the secondary coil and Φ equals the magnetic flux through one turn of the coil. If the turns of the coil are oriented perpendicular to the magnetic field lines, the flux is the product of the magnetic field strength B and the area A through which it cuts. The area is constant, being equal to the cross-sectional area of the transformer core, whereas the magnetic field varies with time according to the excitationof the primary. Since the same magnetic flux passes through both the primary and secondarycoils in an ideal transformer, the instantaneous voltage across the primary winding equals.

$$V_P = N_P \frac{d\Phi}{dt}$$

Taking the ratio of the two equations for V_S and V_P gives the basic equation for stepping up or stepping down the voltage.

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

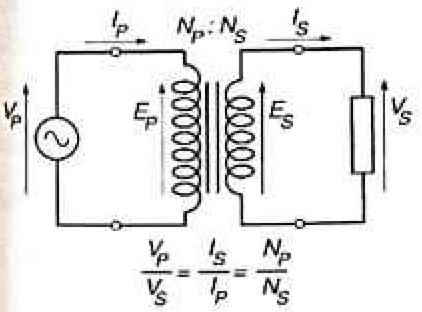
3.8.1.4 : Ideal power equation

If the secondary coil is attached to a load that allows current to flow, electrical power is transmitted from the primary circuit to the secondary circuit. Ideally, the transformer is perfectly efficient; all the incoming energy is transformed from the primary circuit to the magnetic field and into the secondary circuit. If this condition is met, the incoming electric power must equalthe outgoing power.

$$P_{incoming} = I_P V_P = P_{outgoing} = I_S V_S$$

Giving the ideal transformer equation:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$$



$$P_{in\text{-}coming} = I_P V_P = P_{out\text{-}going} = I_S V_S$$

Giving the ideal transformer equation:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$$

If the voltage is increased (stepped up) ($V_S > V_P$), then the current is decreased (stepped down) ($I_S < I_P$) by the same factor. Transformers are efficient so this formula is a reasonable approximation.

If the voltage is increased (stepped up) ($V_S > V_P$), then the current is decreased (stepped down) ($I_S < I_P$) by the same factor. Transformers are efficient so this formula is a reasonable approximation.

The impedance in one circuit is transformed by the *square* of the turns ratio. For example, if an impedance Z_S is attached across the terminals of the secondary coil, it appears to the primary circuit to have an impedance of

$$Z_S\left(\frac{N_P}{N_S}\right)^2$$

This relationship is reciprocal, so that the impedance Z_P of the primary circuit appears to the secondary to be

$$Z_P\left(\frac{N_S}{N_P}\right)^2$$

3.8.1.5 : Detailed operation

The simplified description above neglects several practical factors, in particular theprimary current required to establish a magnetic field in the core, and the contribution to the fielddue to current in the secondary circuit.

Models of an ideal transformer typically assume a core of negligible reluctance with two windings of zero resistance. When a voltage is applied to the primary winding, a small current flows, driving flux around the magnetic circuit of the core. The current required to create the flux is termed the magnetizing current; since the ideal core has been assumed to have near-zero reluctance, the magnetizing current is negligible, although still required to create the magnetic field.

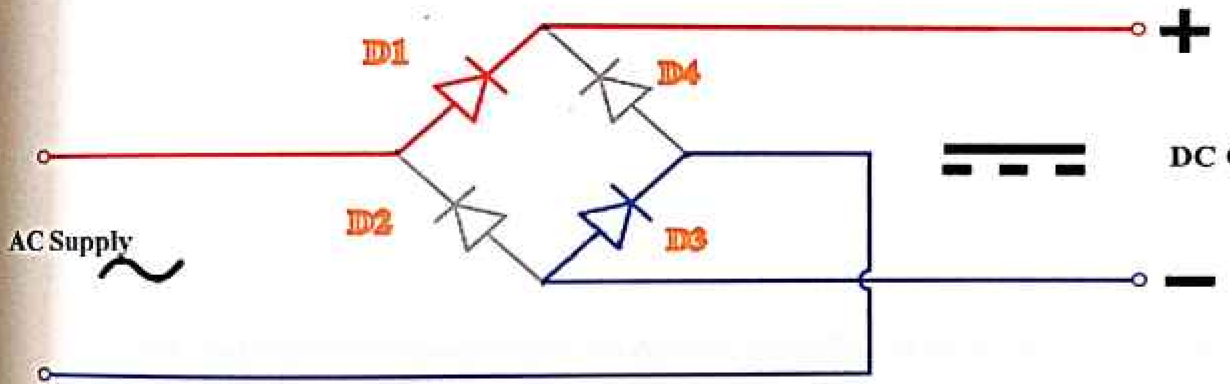
The changing magnetic field induces an electromotive force (EMF) across each winding. Since the ideal windings have no impedance, they have no associated voltage drop, and so the voltages V_P and V_S measured at the terminals of the transformer, are equal to the corresponding EMFs. The primary EMF, acting as it does in opposition to the primary voltage, is sometimes termed the "back EMF". This is due to Lenz's law which states that the induction of EMF would always be such that it will oppose development of any such change in magnetic field.

3.8.1.6 : Bridge Rectifier

A diode bridge or bridge rectifier is an arrangement of four diodes in a bridge configuration that provides the same polarity of output voltage for any polarity of input voltage. When used in its most common application, for conversion of alternating current (AC) input into direct current (DC) output, it is known as a bridge rectifier. A bridge rectifier provides full-wave rectification from a two-wire AC input, resulting in lower cost and weight as compared to a center-tapped transformer design, but has two diode drops rather than one, thus exhibitingreduced efficiency over a center-tapped design for the same output voltage.

3.8.1.7 : Basic Operation

When the input connected at the left corner of the diamond is positive with respect to the one connected at the right hand corner, current flows to the right along the upper coloured path to the output, and returns to the input supply via the lower one.



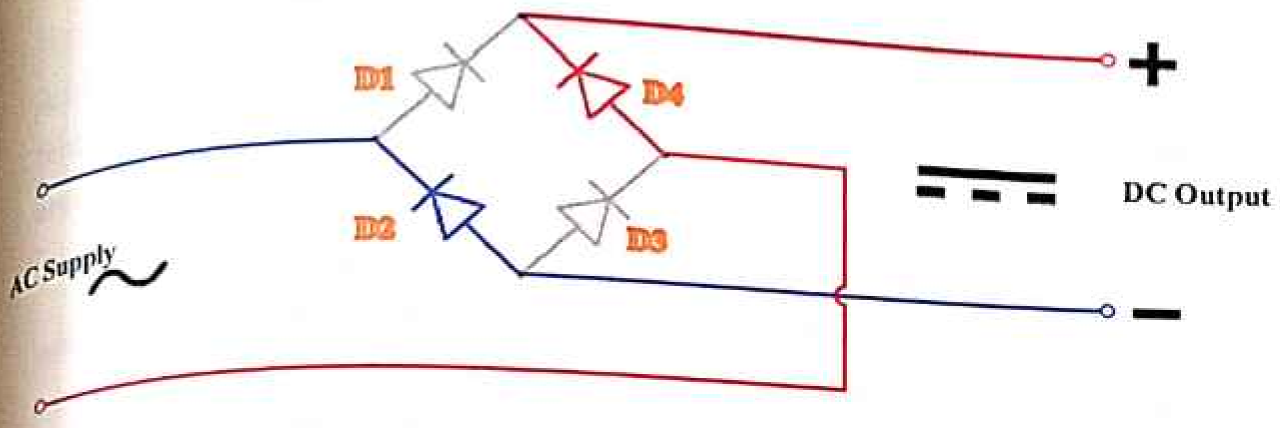


Fig 3.8.1.7: Basic Circuit Diagrams of Bridge Rectifier

In each case, the upper right output remains positive with respect to the lower right one. Since this is true whether the input is AC or DC, this circuit not only produces DC power when supplied with AC power: it also can provide what is sometimes called "reverse polarity protection". That is, it permits normal functioning when batteries are installed backwards or DC input-power supply wiring "has its wires crossed" (and protects the circuitry it powers against damage that might occur without this circuit in place).

Prior to availability of integrated electronics, such a bridge rectifier was always constructed from discrete components. Since about 1950, a single four-terminal component containing the four diodes connected in the bridge configuration became a standard commercial component and is now available with various voltage and current ratings.

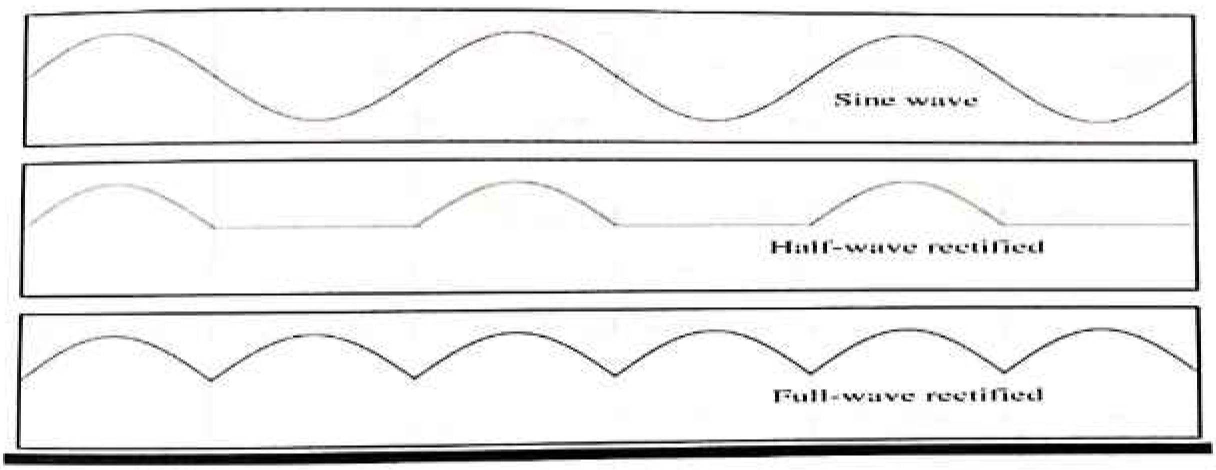


Fig 3.8.1.7.1: Output smoothing (Using Capacitor)

For many applications, especially with single phase AC where the full-wave bridge serves to convert an AC input into a DC output, the addition of a capacitor may be important because the bridge alone supplies an output voltage of fixed polarity but pulsating magnitude(see diagram above).

This bridge rectifier configuration ensures that regardless of the polarity of the input voltage, the output remains consistently positive with respect to the ground or reference point. This not only converts AC power to DC power but also provides a crucial safeguard against damage caused by reverse polarity situations.

By allowing normal operation even when batteries are installed incorrectly or DC power supply wiring is reversed, the circuit effectively protects the connected electronics from potential harm. While historically this rectifier was assembled using discrete components, advancements since around 1950 have led to the development of integrated four-terminal components, streamlining the construction process and offering various voltage and current ratings to suit different applications.

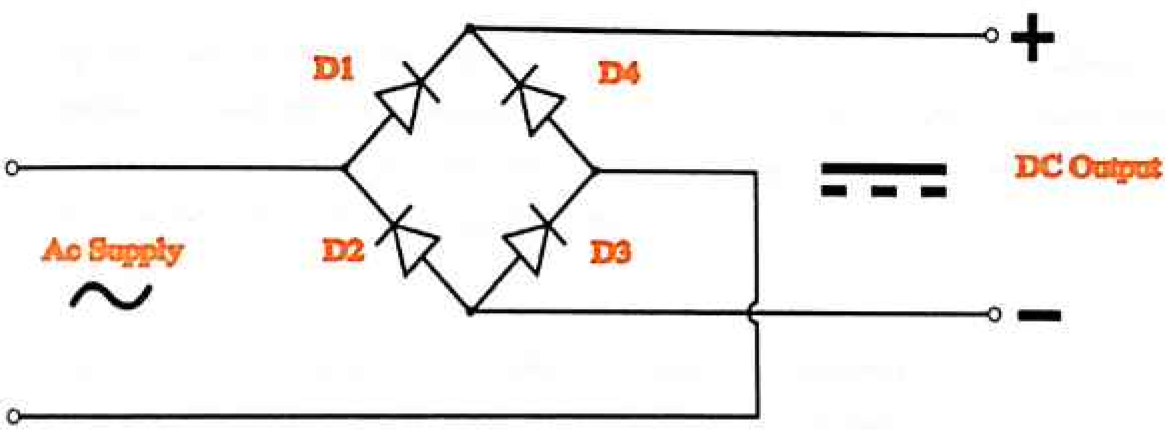


Fig 3.8.1.7.2: Diode Bridge Rectifier

The function of this capacitor, known as a reservoir capacitor (aka smoothing capacitor) is to lessen the variation in (or 'smooth') the rectified AC output voltage waveform from the bridge. One explanation of'smoothing' is that the capacitor provides a low impedance path to the AC component of the output, reducing the AC voltage across, and AC current through, the resistive load. In less technical terms, any drop in the output voltage and current of the bridge tends to be cancelled by loss of charge in the capacitor.

This charge flows out as additional current through the load. Thus the change of load current and voltage is reduced relative to what would occur without the capacitor. Increases of voltage correspondingly store excess charge in the capacitor, thus moderating the change in output voltage / current. Also see rectifier output smoothing.

The simplified circuit shown has a well-deserved reputation for being dangerous, because, in some applications, the capacitor can retain a *lethal* charge after the AC power sourceis removed. If supplying a dangerous voltage, a practical circuit should include a reliable way to safely discharge the capacitor. If the normal load cannot be guaranteed to perform this function, perhaps because it can be disconnected, the circuit should include a bleeder resistor connected asclose as practical across the capacitor. This resistor should consume a current large enough to discharge the capacitor in a reasonable time, but small enough to avoid unnecessary power waste.

Because a bleeder sets a minimum current drain, the regulation of the circuit, defined as percentage voltage change from minimum to maximum load, is improved. However in many cases the improvement is of insignificant magnitude.

The capacitor and the load resistance have a typical time constant $\tau = RC$ where C and R are the capacitance and load resistance respectively. As long as the load resistor is large enough so that this time constant is much longer than the time of one ripple cycle, the above configuration will produce a smoothed DC voltage across the load.

In some designs, a series resistor at the load side of the capacitor is added. The smoothing can then be improved by adding additional stages of capacitor–resistor pairs, often done only for sub-supplies to critical high-gain circuits that tend to be sensitive to supply voltage noise.

The idealized waveforms shown above are seen for both voltage and current when the load on the bridge is resistive. When the load includes a smoothing capacitor, both the voltage and the current waveforms will be greatly changed. While the voltage is smoothed, as described above, current will flow through the bridge only during the time when the input voltage is greater than the capacitor voltage. For example, if the load draws an average current of n Amps, and the diodes conduct for 10% of the time, the average diode current during conduction must be 10n Amps. This non-sinusoidal current leads to harmonic distortion and a poor power factor in the AC supply.

In a practical circuit, when a capacitor is directly connected to the output of a bridge, the bridge diodes must be sized to withstand the current surge that occurs when the power is turned on at the peak of the AC voltage and the capacitor is fully discharged. Sometimes a small series resistor is included before the capacitor to limit this current, though in most applications the power supply transformer's resistance is already sufficient.

Output can also be smoothed using a choke and second capacitor. The choke tends to keep the current (rather than the voltage) more constant. Due to the relatively high cost of an effective choke compared to a resistor and capacitor this is not employed in modern equipment.

Some early console radios created the speaker's constant field with the current from the high voltage ("B +") power supply, which was then routed to the consuming circuits, (permanent magnets were considered too weak for good performance) to create the speaker's constantmagnetic field. The speaker field coil thus performed 2 jobs in one: it acted as a choke, filtering the power supply, and it produced the magnetic field to operate the speaker.

3.8.1.8 : Voltage Regulator

A voltage regulator is an electrical regulator designed to automatically maintain a constant voltage level.

The 78xx (also sometimes known as LM78xx) series of devices is a family of self-contained fixed linear voltage regulator integrated circuits. The 78xx family is a very popular choice for many electronic circuits which require a regulated power supply, due to their ease of use and relative cheapness. When specifying individual ICs within this family, the xx is replaced with a two-digit number, which indicates the output voltage the particular device is designed to provide (for example, the 7805 has a 5 volt output, while the 7812 produces 12 volts). The 78xx line is positive voltage regulators, meaning that they are designed to produce a voltage that is positive relative to a common ground. There is a related line of 79xx devices which are complementary negative voltage regulators. 78xx and 79xx ICs can be used in combination to provide both positive and negative supply voltages in the same circuit, if necessary.78xx ICs have three terminals and are most commonly found in the TO220 form factor, although smaller surface-mount and larger TrO3 packages are also available from some manufacturers. These devices typically support an input voltage which can be anywhere from a couple of volts over theintended output voltage, up to a maximum of 35 or 40 volts, and can typically provide up to around 1 or 1.5 amps of current (though smaller or larger packages may have a lower or higher current rating).

3.9 : LCD

Internal Block Digram

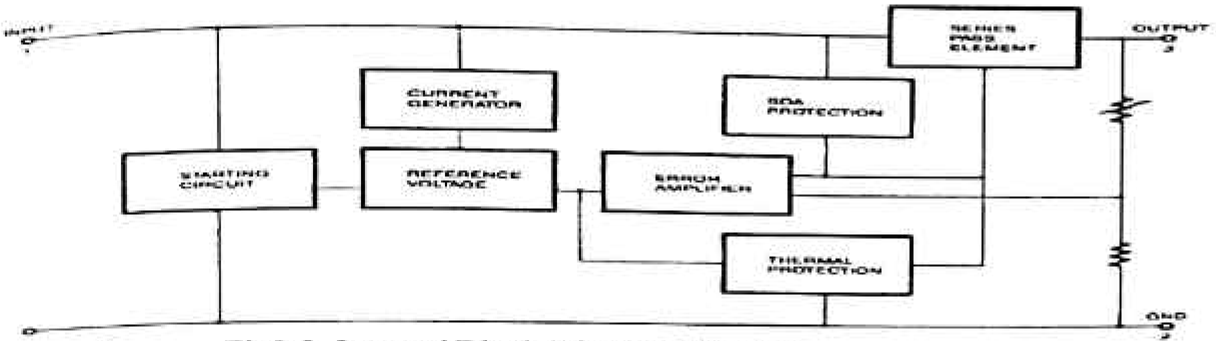


Fig3.9. Internal Block Diagram of LCD

Nowadays, we always use the devices, which are made up of LCDs such as CD players, DVD players, digital watches, computers, etc. These are commonly used in the screen industries to replace the utilization of CRTs. Cathode Ray Tubes use huge power when compared with LCDs, and CRTs heavier as well as bigger. These devices are thinner as well power consumptionis extremely less. The LCD 16×2 working principle is, it blocks the light rather than dissipate. This article discusses of an overview of LCD 16X2, pin configuration and its working.

3.9.1 : What is the LCD 16×2?

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi- segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displayingcustom characters, special and even animations, etc. The LCD 16x2 is a specific type of liquid crystal display module that is widely used in electronic projects and devices. The "16x2" designation refers to the dimensions of the display, indicating that it has 16 character positions in each of its two rows. Each character position can display a single alphanumeric character or symbol.

These modules are popular due to their versatility, low cost, and ease of use. They can be easily interfaced with microcontrollers and other electronic circuits, making them ideal for displaying information in a wide range of applications.

One of the key advantages of the LCD 16x2 module is its programmability. Users can easily control what information is displayed on the screen by sending commands and data to the module via a simple interface. This allows for dynamic content such as sensor readings, messages, and status indicators to be displayed as needed.

Furthermore, the LCD 16x2 module supports the display of custom characters and symbols, providing flexibility for creating unique user interfaces and graphical elements. This capability enables developers to create customized displays tailored to their specific application requirements.

Overall, the LCD 16x2 module is a versatile and cost-effective solution for displaying information in various electronic projects and devices, offering programmability, flexibility, and ease of integration.

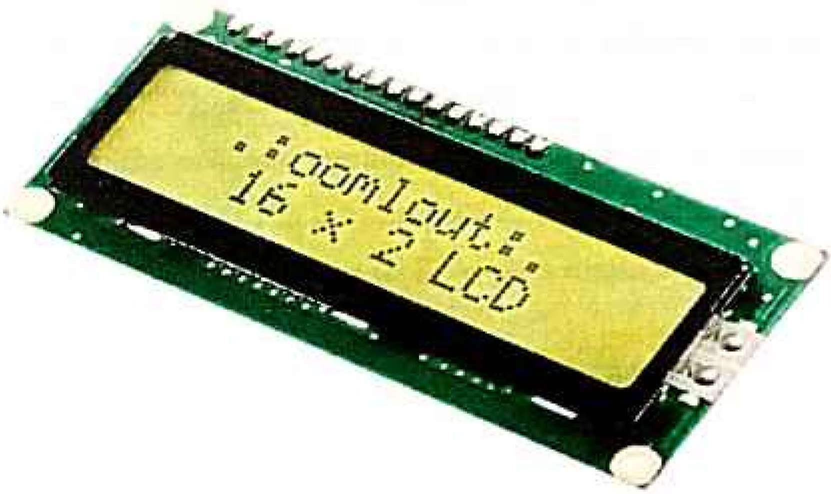


Fig 3.9.1 LCD 16*2

The 16×2 LCD pin out is shown below.

• Pin 1 (Ground/Source Pin): This pin serves as the ground reference for the display. It's crucial to connect it to the ground terminal of the microcontroller or power source to establish a common ground reference for proper operation.

• Pin 2 (VCC/Source Pin): This is the power supply pin for the display module. It typically requires a stable voltage source, such as +5V, to power the display.

• Pin 3 (V0/VEE/Control Pin): This pin is used to adjust the contrast or brightness of the display. By connecting a variable resistor (such as a potentiometer) between this pin and ground, you can control the voltage applied to adjust the display's contrast.

• Pin 4 (Register Select/Control Pin): This pin determines whether the data being sent to the display is interpreted as a command or as display data. When pulled LOW, it indicates data mode,and when pulled HIGH, it indicates command mode.

•Pin 5 (Read/Write/Control Pin): This pin controls the direction of data flow between the microcontroller and the display. When pulled LOW, it indicates a write operation (sending data to the display), and when pulled HIGH, it indicates a read operation (receiving data from the display).

• Pin 6 (Enable/Control Pin): This pin is used to enable the display for data read/write operations. Typically, it needs to be pulsed high to execute these operations. It's often connected to the microcontroller and held high when active.

• Pins 7-14 (Data Pins): These pins are used to transmit data to the display module. The number of pins used depends on the communication mode (4-wire or 8-wire). In 4-wire mode, only four pins (0 to 3) are used for data transfer, while in 8-wire mode, all eight pins (0 to 7) may be utilized.

• Pin 15 (+ve pin of the LED): This pin is typically connected to the positive terminal of the backlight LED, providing the necessary voltage (usually +5V) to illuminate the display.

• Pin 16 (-ve pin of the LED): This pin is connected to the ground terminal of the backlight LED, completing the circuit and allowing current to flow, thus illuminating the display.

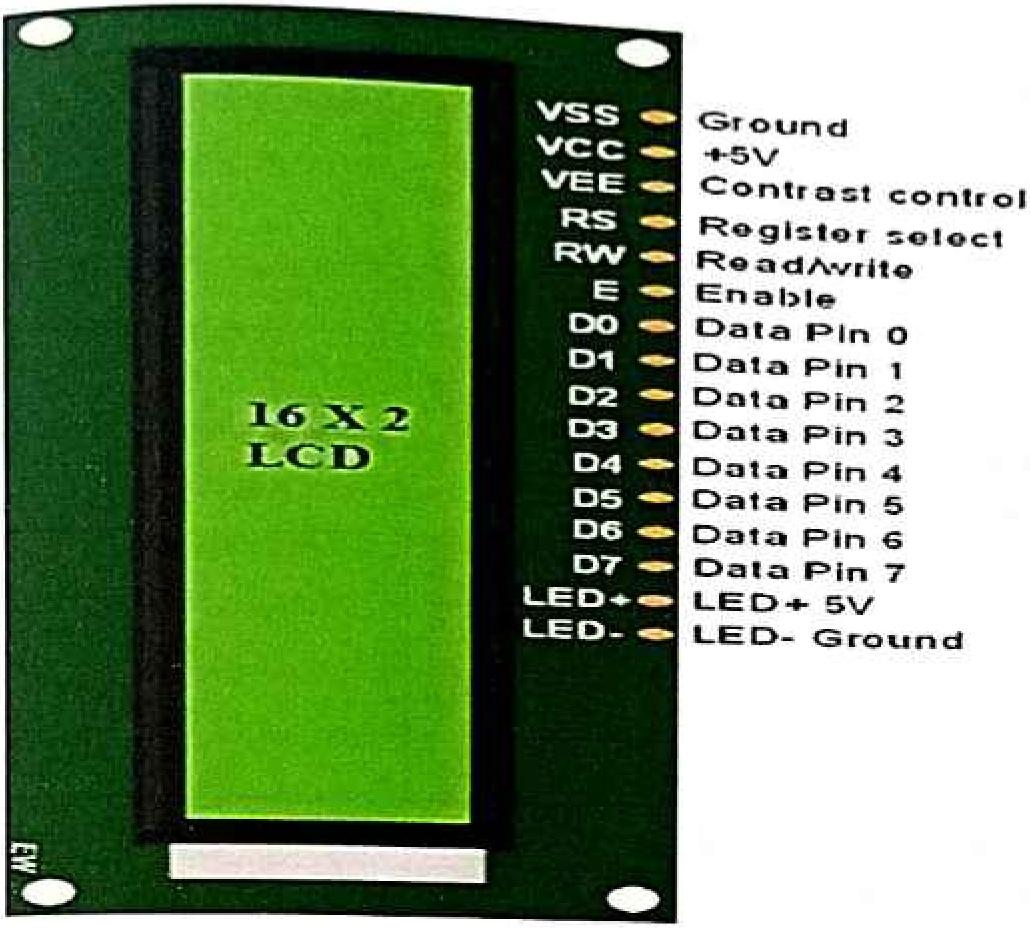


Fig 3.9.2 LCD-16×2-pin-diagram

3.9.2 : Features of LCD16x2

- The features of this LCD mainly include the following.
- The operating voltage of this LCD is 4.7V-5.3V.
- It includes two rows where each row can produce 16-characters.
- The utilization of current is 1mA with no backlight.
- Every character can be built with a 5×8 pixel box.
- The alphanumeric LCDs alphabets & numbers.
- Is display can work on two modes like 4-bit & 8-bit.
- These are obtainable in Blue & Green Backlight.
- It displays a few custom generated characters.

3.9.3 : Registers of LCD

A 16×2 LCD has two registers like data register and command register. The RS (register select) is mainly used to change from one register to another. When the register set is _0‘, then it is known as command register. Similarly, when the register set is _1‘, then it is known as data register.

3.9.4 : Command Register

The main function of the command register is to store the instructions of command which are given to the display. So that predefined tasks can be performed such as clearing the display, initializing, set the cursor place, and display control. Here commands processing can occur within the register.

3.9.5 : Data Register

The main function of the data register is to store the information which is to be exhibited on the LCD screen. Here, the ASCII value of the character is the information which is to be exhibited on the screen of LCD. Whenever we send the information to LCD, it transmits to the data register, and then the process will be starting there. When register set =1, then the data register will be selected.

3.9.6 : 16×2 LCD Commands

The commands of LCD 16X2 include the following.

- For Hex Code-01, the LCD command will be the clear LCD screen
- For Hex Code-02, the LCD command will be returning home
- For Hex Code-04, the LCD command will be decrement cursor
- For Hex Code-06, the LCD command will be Increment cursor
- For Hex Code-05, the LCD command will be Shift display right
- For Hex Code-07, the LCD command will be Shift display left
- For Hex Code-08, the LCD command will be Display off, cursor off
- For Hex Code-0A, the LCD command will be cursor on and display off
- For Hex Code-0C, the LCD command will be cursor off, display on

- For Hex Code-0E, the LCD command will be cursor blinking, Display on
- For Hex Code-0F, the LCD command will be cursor blinking, Display on
- For Hex Code-10, the LCD command will be Shift cursor position to left
- For Hex Code-14, the LCD command will be Shift cursor position to the right
- For Hex Code-18, the LCD command will be Shift the entire display to the left
- For Hex Code-1C, the LCD command will be Shift the entire display to the right
- For Hex Code-80, the LCD command will be Force cursor to the beginning (1st line)
- For Hex Code-C0, the LCD command will be Force cursor to the beginning (2nd line)
- For Hex Code-38, the LCD command will be 2 lines and 5×7 matrix

3.10 : BUZZER

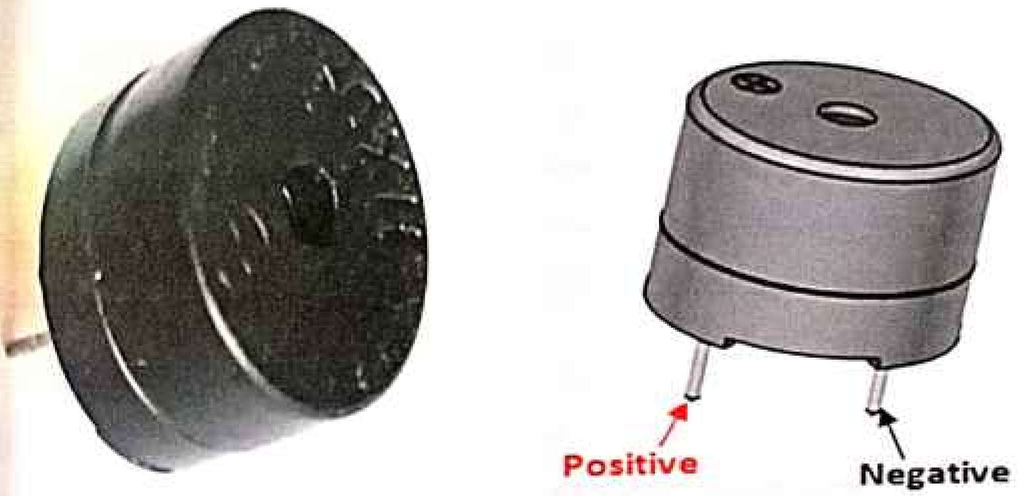


Fig 3.10: Active Passive Buzzer pin out

3.10.1 : Buzzer Pin Configuration

Pin Number: 1

Pin Name: Positive

Description: Identified by (+) symbol or longer terminal lead. Can be powered by DC.

Pin Number: 2

Pin Name: Negative

Description: Identified by short terminal lead. Typically connected to the ground of the circuit.

3.10.2 : Buzzer Features and Specifications

- Rated Voltage: 6V DC
- Operating Voltage: 4-8V DC
- Rated current: <30mA
- Sound Type: Continuous Beep
- Resonant Frequency: ~2300 Hz
- Small and neat sealed package
- Breadboard and Perf board friendly

3.10.3 : How to use a Buzzer

A **buzzer** is a small yet efficient component to add sound features to our project/system. It is very small and compact 2-pin structure hence can be easily used on breadboard, Perf Board and even on PCBs which makes this a widely used component in most electronic applications.

There are two types are buzzers that are commonly available. The one shown here is a simple buzzer which when powered will make a Continuous Beeeeeeppp....sound, the other type is called a readymade buzzer which will look bulkier than this and will produce a Beep. Beep. Beep. Sound due to the internal oscillating circuit present inside it. But, the one shown here is most widely used because it can be customised with help of other circuits to fit easily in our application.

This buzzer can be used by simply powering it using a DC power supply ranging from 4V to 9V. A simple 9V battery can also be used, but it is recommended to use a regulated +5V or +6V DC supply. The buzzer is normally associated with a switching circuit to turn ON or turn OFF the buzzer at required time and require interval.

Moreover, the compact size and low power requirements of the buzzer make it suitable for portable and battery-powered devices. Its compatibility with a wide range of power supplies further enhances its applicability in different scenarios. Whether integrated into a breadboard prototype, soldered onto a perf board, or incorporated directly onto a PCB, the buzzer remains a versatile and essential component in the toolkit of electronics enthusiasts and professionals alike.

3.10.4 : Applications of Buzzer

- Alarming Circuits, where the user has to be alarmed about something
- Communication equipment's
- Automobile electronics
- Portable equipment's, due to its compact size.

3.11 : Temperature Sensor (DHT-11)

Temperature Sensor (DHT-11) is used to monitor temperature and humidity of the atmosphere. The DHT-11 shown in Figure 3 is a basic ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and split out a digital signal on the data pin. The DHT-11 calculate relative humidity by measuring the electrical resistance between two electrodes. The DHT-11 temperature sensor provides a convenient solution for monitoring environmental conditions in various applications, from home automation to industrial systems. Its low cost, compact size, anddigital output make it easy to integrate into electronic projects without requiring complex circuitry or calibration.

By leveraging a combination of a capacitive humidity sensor and a thermistor, the DHT-11 accurately measures both temperature and humidity with reasonable precision. The digital output signal simplifies data acquisition and processing, allowing for straightforward interfacing with microcontrollers and other digital devices.

Additionally, the DHT-11's robust design and relatively wide operating range make it suitable for diverse environments and applications. Whether deployed in indoor climate monitoring systems, greenhouse automation, or weather stations, the DHT-11 temperature sensor offers reliable performance and ease of use, making it a popular choice among hobbyists and professionals alike. The DHT-11 temperature sensor's versatility and affordability make it an excellent choice for various projects. Its ability to accurately measure both temperature and humidity, coupled with its digital output, simplifies integration into electronic systems. Moreover, its robust design ensures reliable performance across different environments.

Furthermore, the DHT-11 can be integrated into weather stations for amateurmeteorology enthusiasts. By collecting data on temperature and humidity, enthusiasts can track weather patterns and trends over time, contributing to local weather monitoring efforts.

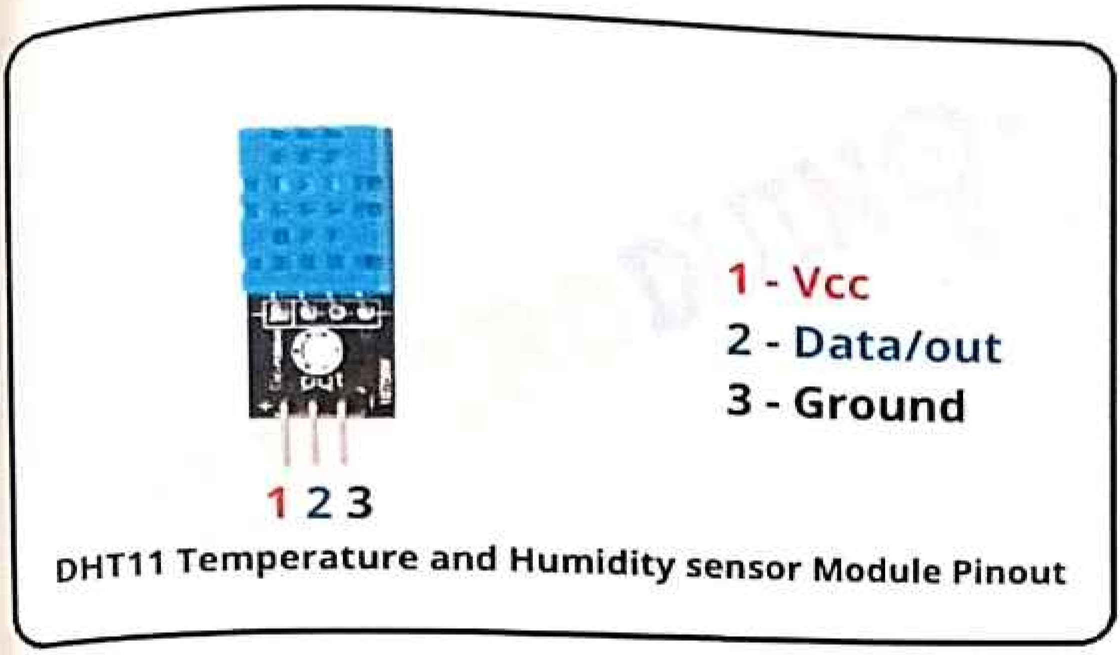


Fig 3.11: DHT11 Sensor

3.12 : WATER PUMP

Water pump The DC 3-6V Mini Micro Submersible Water Pump shown in Figure 5 is a low cost, small size Submersible Pump Motor. It operates with a 2.5 to 6V power supply. It can pump up to 120 litres per hour with a very low current consumption of 220mA. Just connect the tube pipe to the motor outlet, submerge it in water, and power it. The DC 3-6V Mini Micro Submersible Water Pump is a versatile and efficient solution for various water pumping applications. Its compact size and low cost make it suitable for use in small-scale projects and prototypes where space and budget constraints are critical factors.

With a power supply range of 2.5 to 6V, this pump can be easily powered by batteries, low-voltage DC adapters, or renewable energy sources such as solar panels. Its low current consumption of 220mA ensures energy-efficient operation, making it suitable for battery- poweredsystems or continuous operation without significant power consumption.

Capable of pumping up to 120 litres of water per hour, this submersible pump motor is suitable for tasks such as water circulation in aquariums, hydroponic systems, small fountain projects, or even water cooling applications in electronic devices. Its submersible design allows for installation in confined spaces or underwater environments, providing flexibility in installation and usage.

Overall, the DC 3-6V Mini Micro Submersible Water Pump offers reliable performance, easy installation, and cost-effective water pumping solutions for a wide range of applications, making it a popular choice among hobbyists, DIY enthusiasts, and professionals alike.

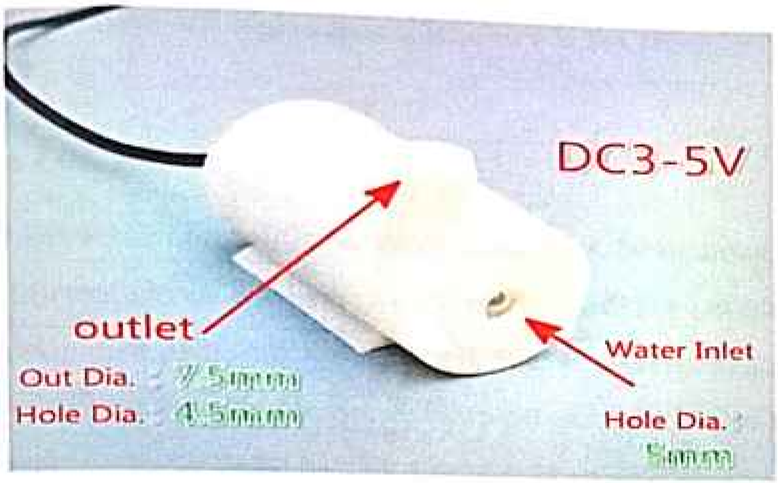


Fig 3.12. Water Pump

3.13 : LDR (Light Dependent Resistor) or Photo resistor

The **Light Dependent Resistor (LDR)** or also popularly known as Photoresistor is just anotherspecial type of Resistor and hence has no polarity so they can be connected in any direction.

They are breadboard friendly and can be easily used on a perf board also. The symbol for LDR is similar to Resistor but includes inward arrows as shown above in the LDR pinout diagram. The arrows indicate the light signals.

3.13.1 : LDR Features:

- Can be used to sense Light
- Easy to use on Breadboard or Perf Board
- Easy to use with Microcontrollers or even with normal Digital/Analog IC
- Small, cheap and easily available
- Available in PG5 ,PG5-MP, PG12, PG12-MP, PG20 and PG20-MP series

3.13.2 : Where to use a LDR:

A **photo resistor** or **LDR** (Light Dependent Resistor), as the name suggests will change it resistance based on the light around it. That is when the resistor is placed in a dark room it will have a resistance of few Mega ohms and as we gradually impose light over the sensor its resistance will start to decrease from Mega Ohms to few Ohms.

This property helps the LDR to be used as a **Light Sensor**. It can detect the amount of light falling on it and thus can predict days and nights. So if you are looking for a sensor to sense light or to distinguish between days and nights then this sensor is the cheap and modest solution for you.

3.13.3 : How to use a LDR Sensor:

As said earlier a LDR is one of the different types of resistors, hence using it is very easy. There are many ways and different circuit in which an LDR can be used. For instance it can be used with Microcontroller Development platforms like Arduino, PIC or even normal Analog IC's like Op-amps. But, here we will use a very simple circuit like a potential divider so that it can be adapted for most of the projects.

A potential Divider is a circuit which has two resistors in series. A constant voltage will be applied across the both the resistor and the output voltage will be measured from the lower resistor. In our case, the lower resistor will be a **LDR** and the constant voltage will be +5V. The set-up is shown below

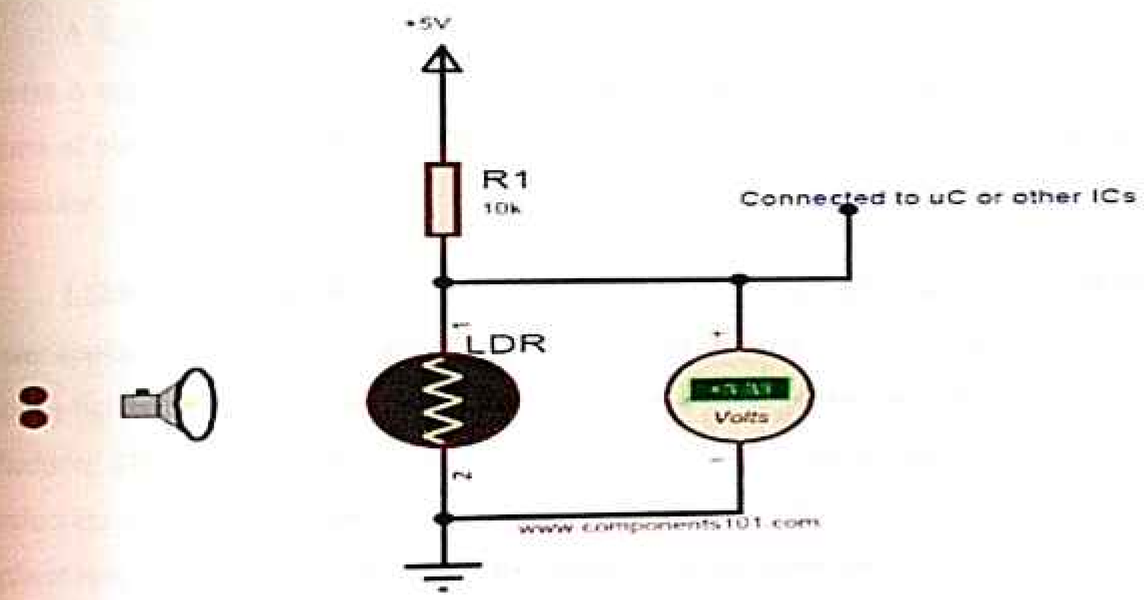


Fig 3.13.3: Light Dependent Resistor

A DC multimeter is used to monitor the voltage across the LDR. As the Lamp is moved towards the resistor the resistance value of the LDR will decrease as a result the voltage drop across it will decrease. The near you bring the Lamp the lower the voltage will get and the farther you move away your Voltage value will increase. The working simulation is shown below.

Now that we have understood how the **LDR provides a variable voltage based on Light**, we can feed this voltage either to a Microcontroller and use ADC to convert the Voltage to meaning full data or use a Op-amp as a Voltage comparator and trigger an specific output like LED or Buzzer for a specific Voltage from the LDR.

3.13.4 : Applications

- Automatic Street Light
- Detect Day or Night
- Automatic Head Light Dimmer
- Position sensor
- Used along with LED as obstacle detector

3.14: LED (LIGHT EMITTING DIODE)

3.14.1: Introduction

A light-emitting diode (LED) is a semiconductor diode that emits light when anelectrical current is applied in the forward direction of the device, as in the simple LED circuit. The effect is a form of electroluminescence. where incoherent and narrow-spectrum light is emitted from the p-n junction.

LEDs are widely used as indicator lights on electronic devices and increasingly in higher power applications such as flashlights and area lighting. An LED is usually a small area (less than 1 mm^2) light source, often with optics added to the chip to shape its radiation pattern and assist in reflection. The colour of the emitted light depends on the composition and condition of the semi conducting material used, and can be infrared, visible, or ultraviolet. Besides lighting, interesting applications include using UV-LEDs for sterilization of water and disinfection of devices, and as a grow light to enhance photosynthesis in plants.

3.14.2 : Basic principle

Like a normal diode, the LED consists of a chip of semi conducting material impregnated, or doped, with impurities to create a p-n junction. As in other diodes, current flows easily from the p-side, or anode, to the n-side, or cathode, but not in the reverse direction. Charge-carriers electrons and holes flow into the junction from electrodes with different voltages. When an electron meets a hole, it falls into a lower energy level, and releases energy in the form of a photon.

The wavelength of the light emitted, and therefore its colour, depends on the band gap energy of the materials forming the p-n junction. In silicon or germanium diodes, the electrons and holes recombine by a non-radiative transition which produces no optical emission, because these are indirect band gap materials. The materials used for the LED have a direct band gap with energies corresponding to near-infrared, visible or near-ultraviolet light. LED development began with infrared and red devices made with gallium arsenide. Advances in materials science have made possible the production of devices with ever-shorter wavelengths, producing light in avariety of colours. LEDs are usually built on an n-type substrate, with an electrode attached to the p-type layer deposited on its surface. P-type substrates, while less common, occur as well. Many commercial LEDs, especially GaN/InGaN, also use sapphire substrate.

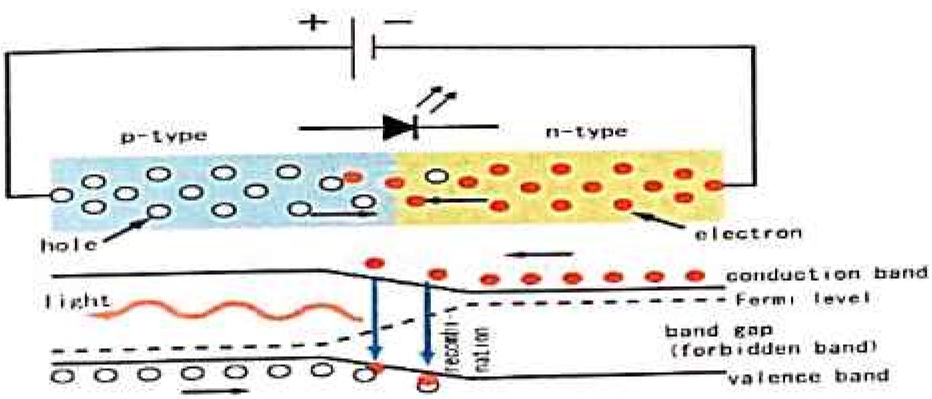


Fig 3.14.2.Basic Circuit Diagram of LED

3.14.3 LED Display types:

- Bar graph
- Seven segment
- Star burst
- Dot matrix

3.14.4 : Basic LED types

3.14.4.1: Miniature LEDs

Fig 3.14.4.1: Miniature LED

Different sized LEDs. 8 mm, 5mm and 3 mm.

These are mostly single-die LEDs used as indicators, and they come in various-size packages:

- surface mount
- 2 mm
- 3 mm (T1)
- 5 mm (T1¾)
- 10 mm
- Other sizes are also available, but less common.

Common package shapes:

- Round, dome top
- Round, flat top
- Rectangular, flat top (often seen in LED bar-graph displays)
- Triangular or square, flat top

The encapsulation may also be clear or semi opaque to improve contrast and viewing angle.

There are three main categories of miniature single die LEDs:

- Low current — typically rated for 2 mA at around 2 V (approximately 4 mW consumption).
- Standard — 20 mA LEDs at around 2 V (approximately 40 mW) for red, orange, yellow & green, and 20 mA at 4–5 V (approximately 100 mW) for blue, violet and white.
- Ultra-high output — 20 mA at approximately 2 V or 4–5 V, designed for viewing in direct sunlight.

3.14.5 : Five- and twelve-volt LEDs

These are miniature LEDs incorporating a series resistor, and may be connected directly to a 5 V or 12 V supply.

3.14.5.1 : Flashing LEDs

Flashing LEDs are used as attention seeking indicators where it is desired to avoidthe complexity of external electronics. Flashing LEDs resemble standard LEDs but they contain an integrated multi vibrator circuit inside which causes the LED to flash with a typical period of one second. In diffused lens LEDs this is visible as a small black dot. Most flashing LEDs emit light of a single colour, but more sophisticated devices can flash between multiple colours and even fade through a colour sequence using RGB colour mixing.

3.14.5.2 : High power LEDs

High power LEDs from lumi leds mounted on a star shaped heat sink High power LEDs (HPLED) can be driven at more than one ampere of current and give out large amounts of light. Since overheating destroys any LED the HPLEDs must be highly efficient to minimize excess heat, furthermore they are often mounted on a heat sink to allow for heat dissipation. If the heat from a HPLED is not removed the device will burn out in seconds.

A single HPLED can often replace an incandescent bulb in a flashlight or be set in an array to form a powerful LED lamp. LEDs have been developed that can run directly from mainspower without the need for a DC converter. For each half cycle part ofthe LED diode emits lightand part is dark, and this is reversed during the next half cycle. Current efficiency is 80 lm/W.

3.14.5.3 : Multi-colour LEDs

A -bi-colour LEDl is actually two different LEDs in one case. It consists of two dies connected to the same two leads but in opposite directions. Current flow in one direction produces one colour, and current in the opposite direction produces the other colour. Alternating the two colours with sufficient frequency causes the appearance of a third colour. A -tri-colour LEDl is also two LEDs in one case, but the two LEDs are connected to separate leads so that the two LEDs can be controlled independently and lit simultaneously.

RGB LEDs contain red, green and blue emitters, generally using a four-wire connection with one common (anode or cathode). The Taiwanese LED manufacturer Ever light has introduced a 3watt RGB package capable of driving each die at 1 watt.

3.14.5.4 : Alphanumeric LEDs

LED displays are available in seven-segment and starburst format. Seven-segment displays handle all numbers and a limited set of letters. Starburst displays can display all letters. Seven-segment LED displays were in widespread use in the 1970s and 1980s, but increasing use of liquid crystal displays, with their lower power consumption and greater display flexibility, has reduced the popularity of numeric and alphanumeric LED displays.

3.14.5.5 : Applications

- **Automotive applications with LEDS**

Instrument Panels & Switches, Courtesy Lighting, CHMSL, Rear Stop/Turn/Tai, Retrofits, New Turn/Tail/Marker Lights.

- **Consumer electronics & general indication**

Household appliances, VCR/ DVD/ Stereo/Audio/Video devices, Toys/Games Instrumentation, Security Equipment, Switches.

- **Illumination with LEDs**

Architectural Lighting, Signage (Channel Letters), Machine Vision, Retail Displays, Emergency Lighting (Exit Signs), Neon and bulb Replacement, Flashlights, Accent Lighting - Pathways, Marker Lights.

- **Sign applications with LEDs**

Full Colour Video, Monochrome Message Boards, Traffic/VMS, Transportation Passenger Information.

- **Signal application with LEDs**

Traffic, Rail, Aviation, Tower Lights, Runway Lights, Emergency/Police Vehicle Lighting.

- **Mobile applications with LEDs**

Mobile Phone, PDA's, Digital Cameras, Lap Tops, General Backlighting.

- **Photo sensor applications with LEDs**

Medical Instrumentation, Bar Code Readers, Colour & Money Sensors, Encoders, Optical Switches, Fibre Optic Communication.

Fig 3.14.5 Five and twelve volt LED

3.15 : ULTRASONIC SENSOR

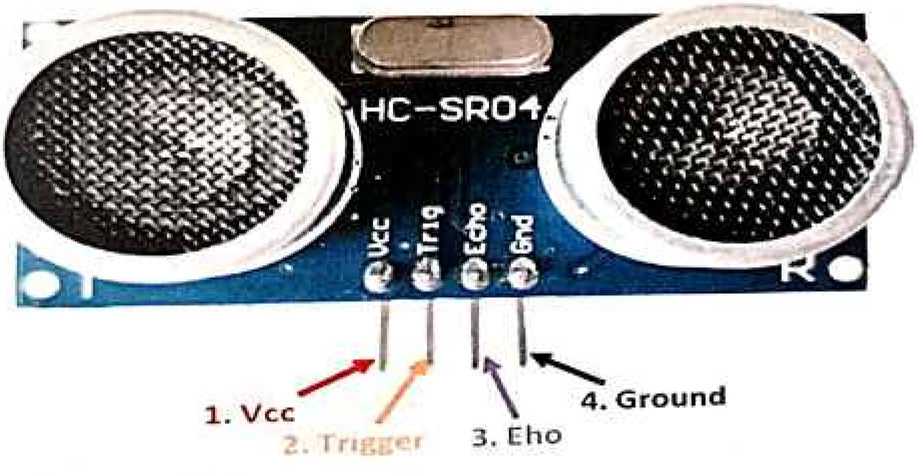


Fig 3.15.Ultrasonic Sensor HC-SR04

3.15.1 : HC-SR04 Sensor Features

- Operating voltage: +5V
- Theoretical Measuring Distance: 2cm to 450cm
- Practical Measuring Distance: 2cm to 80cm
- Accuracy: 3mm
- Measuring angle covered: <15°
- Operating Current: <15mA
- Operating Frequency: 40Hz

3.15.2 : HC-SR04 Ultrasonic Sensor - Working

As shown above the HC-SR04 Ultrasonic (US) sensor is a 4 pin module, whose pin names are Vcc, Trigger, Echo and Ground respectively. This sensor is a very popular sensor used in many applications where measuring distance or sensing objects are required. The module has two eyes like projects in the front which forms the Ultrasonic transmitter and Receiver. The sensor works with the simple high school formula that

$$\textbf{Distance} = \textbf{Speed} \times \textbf{Time}$$

The Ultrasonic transmitter transmits an ultrasonic wave, this wave travels in air and when it gets objected by any material it gets reflected back toward the sensor this reflected wave is observed by the Ultrasonic receiver module as shown in the picture below

Fig3.15.1. Working of HC-SR04 Ultrasonic Sensor

Now, to calculate the distance using the above formulae, we should know the Speed and time. Since we are using the Ultrasonic wave we know the universal speed of US wave at room conditions which is 330m/s. The circuitry inbuilt on the module will calculate the time taken for the US wave to come back and turns on the echo pin high for that same particular amount of time, this way we can also know the time taken. Now simply calculate the distance using a microcontroller or microprocessor.

3.15.3 : How to use the HC-SR04 Ultrasonic Sensor

HC-SR04 distance sensor is commonly used with both microcontroller device used and microprocessor platforms like Arduino, ARM, PIC, Raspberry Pie etc. The following guide is universally since it has to be followed irrespective ofthe type of computational device used.

Power the Sensor using a regulated +5V through the Vcc ad Ground pins ofthe sensor. The current consumed by the sensor is less than 15mA and hence can be directly powered by the on board 5V pins (If available). The Trigger and the Echo pins are both I/O pins and hence they can be connected to I/O pins of the microcontroller. To start the measurement, the trigger pin hasto be made high for 10uS and then turned off. This action will trigger an ultrasonic wave at frequency of 40Hz from the transmitter and the receiver will wait for the wave to return. Once the wave is returned after it getting reflected by any object the Echo pin goes high for a particular amount of time which will be equal to the time taken for the wave to return back to thesensor.

The amount of time during which the Echo pin stays high is measured by the MCU/MPU as it gives the information about the time taken for the wave to return back to the Sensor. Using this information the distance is measured as explained in the above heading.

Applications:

- Used to avoid and detect obstacles with robots like biped robot, obstacle avoider robot, path finding robot etc.
- Used to measure the distance within a wide range of 2cm to 400cm
- Can be used to map the objects surrounding the sensor by rotating it
- Depth of certain places like wells, pits etc can be measured since the waves can penetrate through water.

3.16 : Solar panel

A 12V solar panel is a photovoltaic module designed to generate electricity from sunlight, typically producing a nominal voltage of 12 volts. These panels consist of multiple solar cells connected in series to achieve the desired voltage output.

The 12V designation refers to the nominal voltage output of the panel when operating under standard test conditions (STC), which typically include a specific level of sunlight intensity and temperature. In practical applications, the actual voltage output may vary depending on factorssuch as sunlight intensity, angle of incidence, and temperature.

These solar panels are commonly used in off-grid and grid-tied solar power systems for various applications, including:

1. Off-grid power systems: 12V solar panels are often used in off-grid applications such as cabins, RVs, boats, and remote homes where grid electricity is not available. They charge batteries to store energy for use when sunlight is not available.

2. Solar water pumping: 12V solar panels can power water pumps directly or charge batteries for water pumping systems used in irrigation, livestock watering, and remote water supply applications.

3. Outdoor lighting: These panels are used to power outdoor lighting systems, including garden lights, pathway lights, and security lights, providing illumination without the need for grid electricity.

4. Portable power solutions: 12V solar panels are also used in portable power solutions such as solar chargers for electronic devices, portable solar generators, and camping kits, providing renewable energy for outdoor activities and emergencies.

When selecting a 12V solar panel, factors to consider include the panel's wattage, efficiency, durability, and compatibility with charge controllers and other system components. Additionally, proper installation and maintenance are essential for maximizing the performance and lifespan of the solar panel system.

Fig 3.16. Solar Panel

3.17 : Charge Controller

A solar charge controller, also known as a solar regulator, is a critical component in a solar power system that regulates the voltage and current from solar panels to batteries or loads. Its primary function is to ensure the proper charging of batteries and protect them from overcharging, over-discharging, and other potential issues. Here's an overview of the functions and features of a solar charge controller:

1. Voltage Regulation: Solar panels produce varying voltage levels depending on factors such as sunlight intensity and temperature. The charge controller regulates this voltage to ensure that it matches the requirements of the battery bank or load.

2. Current Regulation: The charge controller limits the current flowing from the solar panels to prevent overcharging of the batteries. It also regulates the current flowing to loads connected to the system.

3. Battery Protection: Charge controllers monitor the state of charge (SOC) of the batteries and implement charging algorithms to optimize battery performance and lifespan. They protect batteries from overcharging by reducing or stopping the charging current when the batteries reach full capacity.

4. Load Control: Some charge controllers include load terminals to connect and control DC loads such as lights, fans, or other electrical devices. They can automatically disconnect the load from the batteries when the battery voltage drops to a certain level to prevent over-discharging.

5. Temperature Compensation: Charge controllers may include temperature sensors to adjust the charging voltage based on the temperature of the batteries, ensuring optimal charging efficiency and battery longevity.

6. Display and Monitoring: Many modern charge controllers feature built-in displays or communication ports that allow users to monitor system performance, battery status, and other parameters in real-time.

7. Multiple Charging Modes: Advanced charge controllers may offer multiple charging modes such as bulk, absorption, float, and equalization charging, providing flexibility to optimize charging based on battery chemistry and usage patterns.

When selecting a solar charge controller, factors to consider include the maximum solar panel and battery voltage ratings, maximum charging current, efficiency, temperature compensation capabilities, and additional features such as load control and monitoring options. Proper sizing and installation of the charge controller are crucial to ensure the safety and efficiency of the solar power system.

Fig 3.17. Charge Controller

3.18 : Batteries

A 3.7V lithium-ion (Li-ion) battery is a rechargeable battery commonly used in various electronic devices due to its high energy density, relatively light weight, and long cycle life. These batteries utilize lithium ions as the charge carriers during discharge and recharge cycles. Here are some key features and considerations regarding 3.7V lithium-ion batteries:

1. Nominal Voltage: The nominal voltage of a lithium-ion battery is typically 3.7 volts. This voltage remains relatively stable throughout most of the discharge cycle, providing consistent power to devices.

2. Energy Density: Lithium-ion batteries offer a high energy density compared to other rechargeable batteries, allowing them to store a significant amount of energy in a relatively compact and lightweight form.

3. Recharge ability: One of the primary advantages of lithium-ion batteries is their recharge ability. They can be recharged hundreds to thousands of times without significant loss of capacity, making them cost-effective and environmentally friendly.

4. Voltage Range: While the nominal voltage of a lithium-ion battery is 3.7 volts, the actual voltage can vary depending on the state of charge. When fully charged, the voltage can reach around 4.2 volts, and when fully discharged, it can drop to around 3.0 volts.

5. Protection Circuitry: To prevent overcharging, over-discharging, and short circuits, lithium- ion batteries often include built-in protection circuitry. This circuitry helps ensure safe and reliable operation of the battery in various applications.

7. Charging Considerations: It's crucial to use a compatible charger specifically designed for lithium-ion batteries to ensure safe and efficient charging. Overcharging or charging at incorrect voltages can lead to safety hazards and reduce battery lifespan.

Overall, 3.7V lithium-ion batteries offer a reliable and versatile power source for a wide range of consumer electronics, providing long-lasting performance and durability.

Fig 3.18. Batteries

CHAPTER-4

INTRODUCTION TO ARDUINO UNO SOFTWARE

4.1 : Software Description

4.1.1 Arduino IDE:

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

4.1.2: Writing Sketches

Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

NB: Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension .pde. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the .ino extension on save.

Verify
Checks your code for errors compiling it.

Upload
Compiles your code and uploads it to the configured board. See uploading below details.

Note: If you are using an external programmer with your board, you can hold down "shift" key on your computer when using this icon. The text will change to "Upl using Programmer"

New
Creates a new sketch.

Open
Presents a menu of all the sketches in your sketchbook. Clicking one will open it wit the current window overwriting its content.

Note: due to a bug in Java, this menu doesn't scroll; if you need to open a sketch lat the list, use the **File | Sketch book** menu instead.

Save
Saves your sketch.

SerialMonitor
Opens the serial monitor.

Additional commands are found within the five menus: **File, Edit, Sketch, Tools, Help**. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

4.1.2.1 : File

- New

  Creates a new instance of the editor, with the bare minimum structure of a sketch already in place.
- Open

  Allows to load a sketch file browsing through the computer drives and folders.
- Openrecent

  Provides a short list of the most recent sketches, ready to be opened.

- Sketchbook

  Shows the current sketches within the sketchbook folder structure; clicking on any name opens the corresponding sketch in a new editor instance.

- Examples

  Any example provided by the Arduino Software (IDE) or library shows up in this menu item. All the examples are structured in a tree that allows easy access by topic or library.

- Close

  Closes the instance of the Arduino Software from which it is clicked.

- Save

  Saves the sketch with the current name. If the file hasn't been named before, a name will be provided in a "Save as.." window.

- Saveas...

  Allows to save the current sketch with a different name.

- PageSetup

  It shows the Page Setup window for printing.

- Print

  Sends the current sketch to the printer according to the settings defined in Page Setup.

- Preferences

  Opens the Preferences window where some settings of the IDE may be customized, as the language of the IDE interface.

- Quit

  Closes all IDE windows. The same sketches open when Quit was chosen will be automatically reopened the next time you start the IDE.

4.1.2.2 : Edit

- Undo/Redo

  Goes back of one or more steps you did while editing; when you go back, you may go forward with Redo.

- Cut

  Removes the selected text from the editor and places it into the clipboard.

- Copy

  Duplicates the selected text in the editor and places it into the clipboard.

- CopyforForum

  Copies the code of your sketch to the clipboard in a form suitable for posting to the forum, complete with syntax colouring.
- CopyasHTML

  Copies the code of your sketch to the clipboard as HTML, suitable for embedding in web pages.
- Paste

  Puts the contents of the clipboard at the cursor position, in the editor.
- SelectAll

  Selects and highlights the whole content of the editor.
- Comment/Uncomment

  Puts or removes the // comment marker at the beginning of each selected line.
- Increase/DecreaseIndent

  Adds or subtracts a space at the beginning of each selected line, moving the text one space on the right or eliminating a space at the beginning.
- Find

  Opens the Find and Replace window where you can specify text to search inside the current sketch according to several options.
- FindNext

  Highlights the next occurrence - if any - of the string specified as the search item in the Find window, relative to the cursor position.
- FindPrevious

  Highlights the previous occurrence - if any - of the string specified as the search item in the Find window relative to the cursor position.
- Verify/Compile

  Checks your sketch for errors compiling it; it will report memory usage for code and variables in the console area.
- Upload

  Compiles and loads the binary file onto the configured board through the configured Port.

- UploadUsingProgrammer

  This will overwrite the bootloader on the board; you will need to use Tools > Burn Bootloader to restore it and be able to Upload to USB serial port again. However, it allows you to use the full capacity of the Flash memory for your sketch. Please note that this command will NOT burn the fuses. To do so a *Tools -> Burn Bootloader* command must be executed.
- ExportCompiledBinary

  Saves a .hex file that may be kept as archive or sent to the board using other tools.
- ShowSketchFolder

  Opens the current sketch folder.
- IncludeLibrary

  Adds a library to your sketch by inserting #include statements at the start of your code. For more details, see libraries below. Additionally, from this menu item you can access the Library Manager and import new libraries from .zip files.
- AddFile...

  Adds a source file to the sketch (it will be copied from its current location). The new file appears in a new tab in the sketch window. Files can be removed from the sketch using the tab menu accessible clicking on the small triangle icon below the serial monitor one on the right side o the toolbar.

4.1.2.3 : Tools

- AutoFormat

  This formats your code nicely: i.e. indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more.
- ArchiveSketch

  Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.
- FixEncoding&Reload

  Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
- SerialMonitor

  Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.

- Board

  Select the board that you're using. See below for descriptions of the various boards.

- Port

  This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu.

- Programmer

  For selecting a harware programmer when programming a board or chip and not using the onboard USB-serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this.

- Burn *Bootloader*

  The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino or Genuino board but is useful if you purchase a new ATmega microcontroller (which normally come without a bootloader). Ensure that you've selected the correct board from the **Boards** menu before burning the bootloader on the target board. This command also set the right fuses.

4.1.2.4 : Help

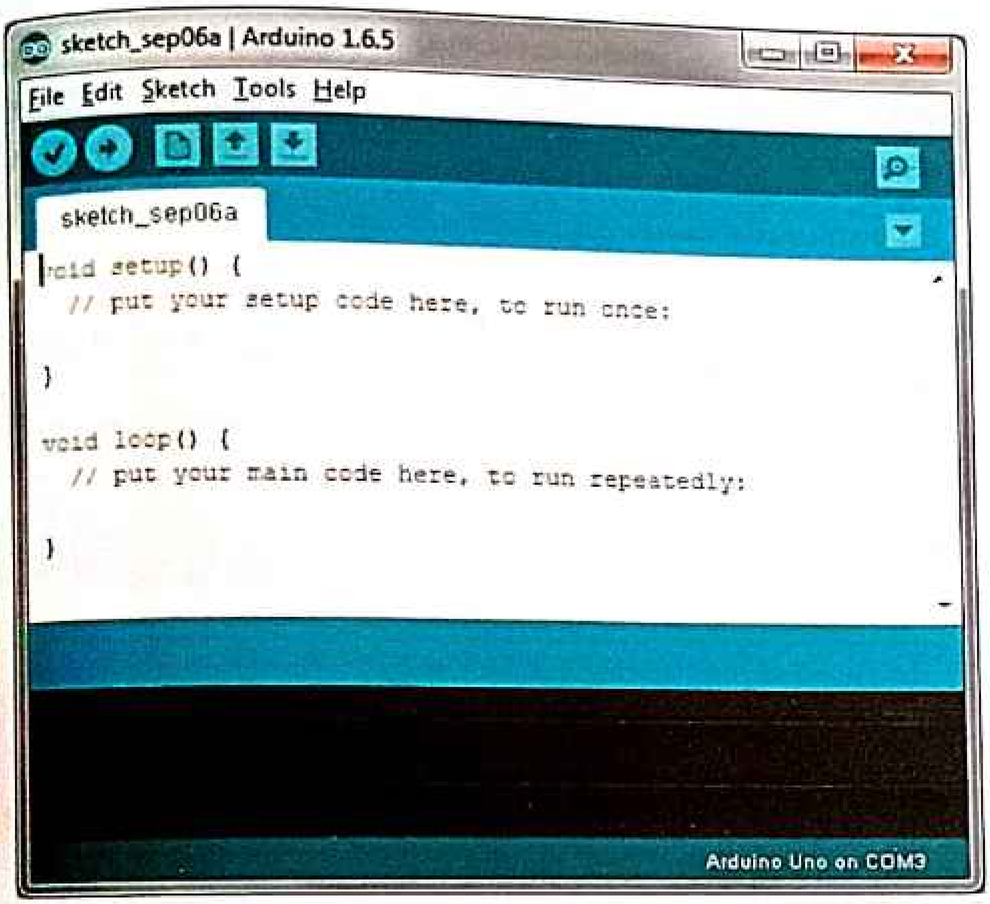
Here you find easy access to a number of documents that come with the Arduino Software (IDE). You have access to Getting Started, Reference, this guide to the IDE and other documents locally, without an internet connection. The documents are a local copy of the online ones and may link back to our online website.

- FindinReference

  This is the only interactive function of the Help menu: it directly selects the relevant page in the local copy of the Reference for the function or command under the cursor.

4.2: Arduino IDE: Initial Setup

This is the Arduino IDE once it's been opened. It opens into a blank sketch where you can start programming immediately. First, we should configure the board and port settings to allow us to upload code. Connect your Arduino board to the PC via the USB cable.

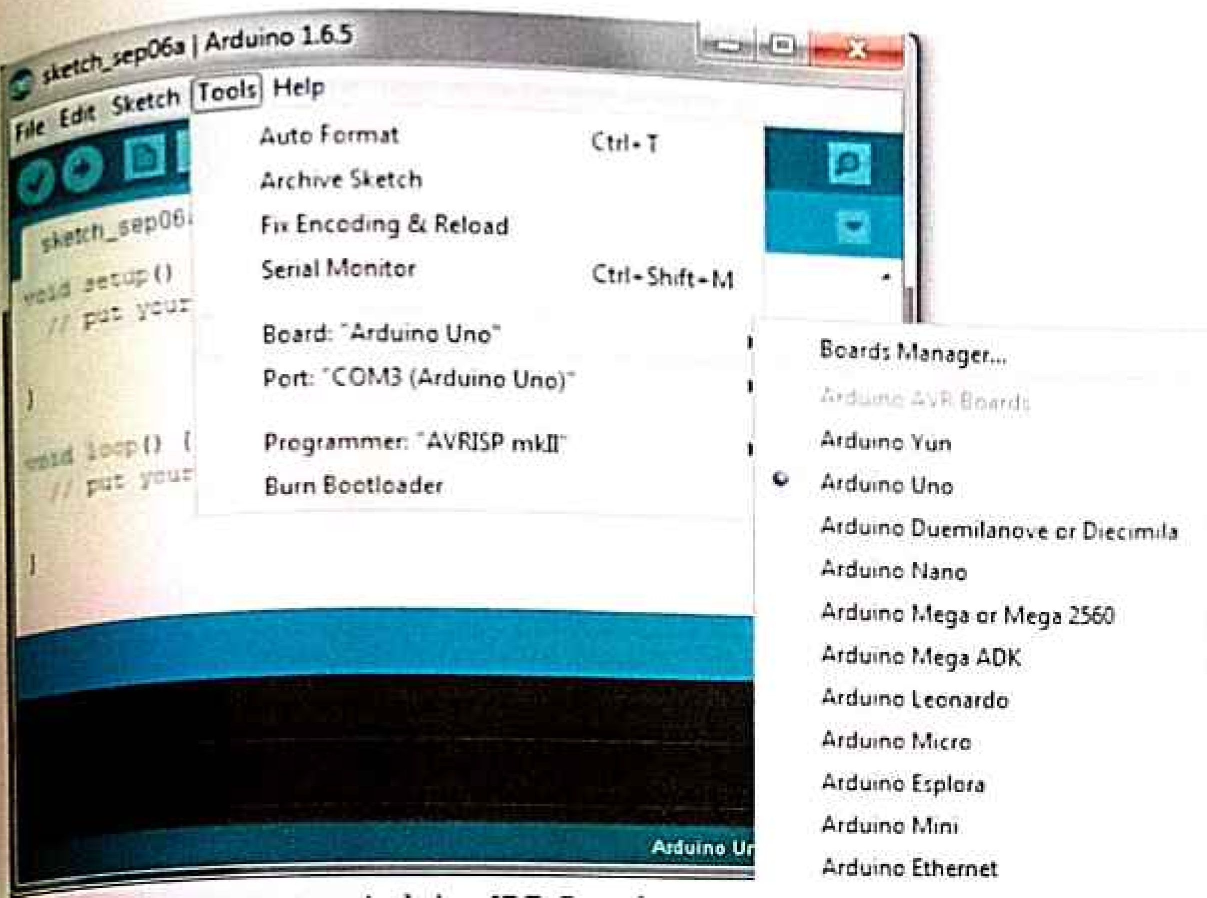


Arduino IDE Default Window

Fig 4.2 Initial setup

4.2.1. IDE: Board Setup

You have to tell the Arduino IDE what board you are uploading to. Select the Toolspulldown menu and go to Board.This list is populated by default with the currently available Arduino Boards that are developed by Arduino. If you are using an Uno or an Uno- Compatible Clone (ex. Funduino, SainSmart, IEIK, etc.), select Arduino Uno. If you are using another board/clone, select that board.

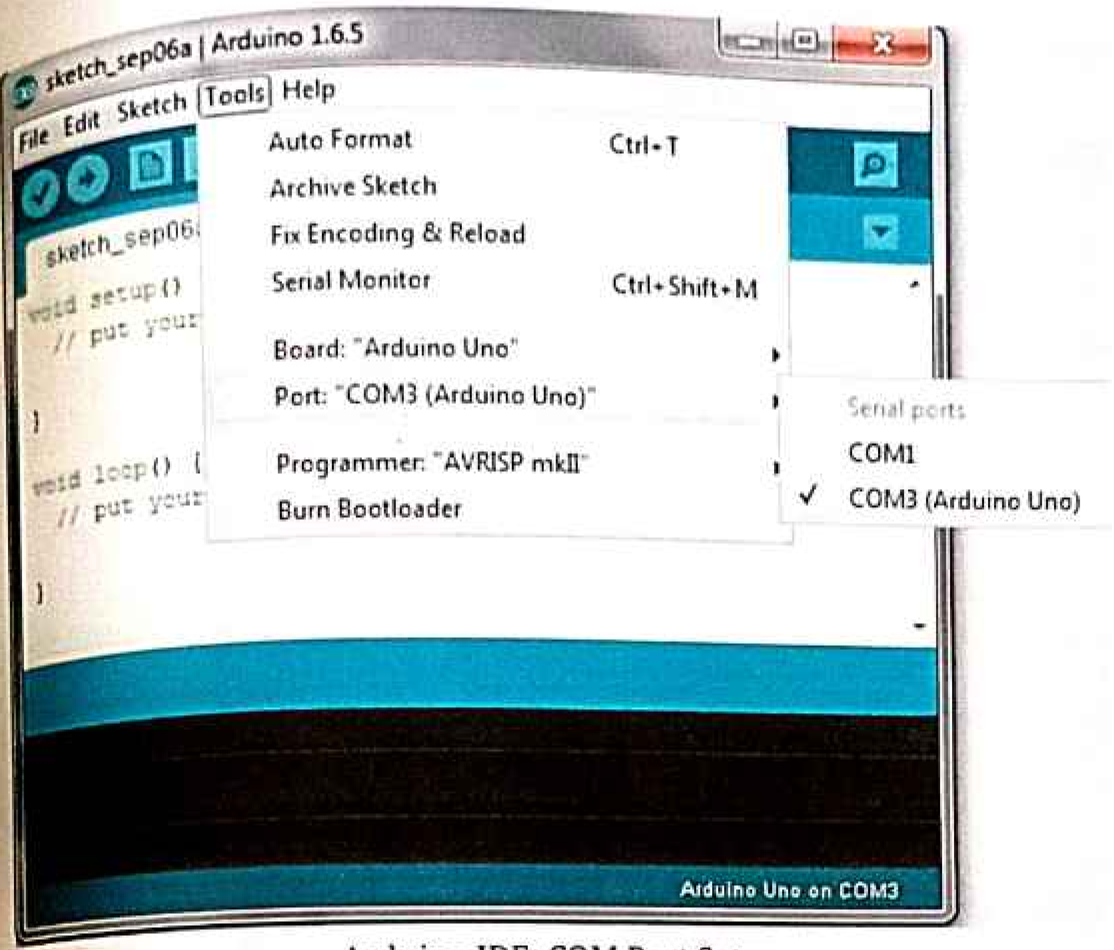


Arduino IDE: Board Setup Procedure

Fig 4.2.1.Board setup

4.2.2 IDE: COM Port Setup

If you downloaded the Arduino IDE before plugging in your Arduino board, when you plugged in the board, the USB drivers should have installed automatically. The most recent Arduino IDE should recognize connected boards and label them with which COM port they are using. Select the Tools pulldown menu and then Port.Here it should list all open COM ports, and if there is a recognized Arduino Board, it will also give it's name. Select the Arduino board that you have connected to the PC. If the setup was successful, in the bottom right of the Arduino IDE, you should see the board type and COM number of the board you plan to program. Note: the Arduino Uno occupies the next available COM port; it will not always be COM3.



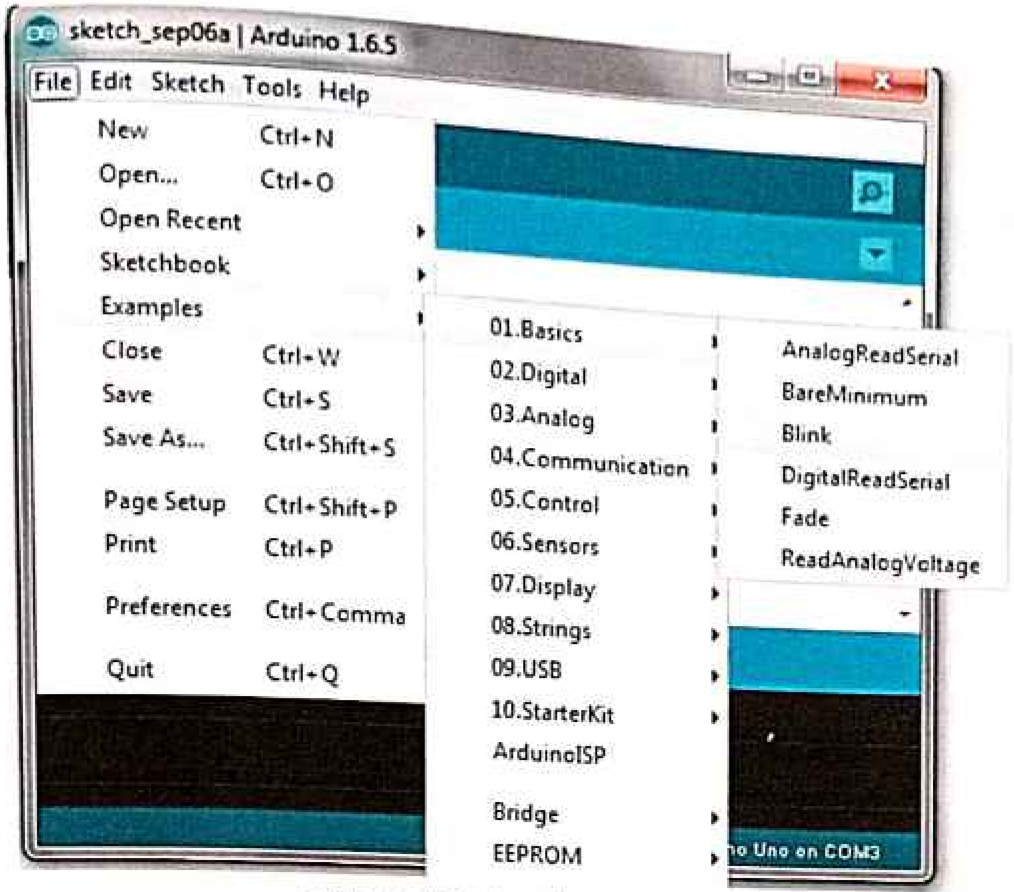
Arduino IDE: COM Port Setup

Fig 4.2.2 Com port setup

4.2.3. Testing Your Settings:

Uploading Blink One common procedure to test whether the board you are using is properly set up is to upload the –Blink‖ sketch. This sketch is included with all Arduino IDE releases and can be accessed by the Filepull-down menu and going to Examples, 01.Basics, and then select Blink. Standard Arduino Boards include a surface-mounted LED labeled –L‖ or –LED‖ next to the –RX‖ and –TX‖ LEDs, that is connected to digital pin 13. This sketch will blink the LED at a regular interval, and is an easy way to confirm if your board is set up properly and you were successful in uploading code. Open the –Blink‖ sketch and press the –Upload‖ button in the upper-left corner to upload –Blink‖ to the board.

Upload Button:



Arduino IDE: Loading Blink Sketch

Fig4.2.3 Loading Blink Sketch

4.3: Thing Speak IoT Platform

Thing Speak is IoT platform for user to gather real-time data; for instance, climate information, location data and other device data. In different channels in Thing Speak, you can summarize information and visualize data online in charts and analyse useful information.Thing Speak can integrate IoT: bit (micro:bit) and other software/ hardware platforms. Through IoT: bit, you can upload sensors data to Thing Speak (e.g. temperature, humidity, light intensity, noise, motion, raindrop, distance and other device information).

Fig 4.3 Thing speak Configuration

Goal:

we need to create the thing speak channel and get the key

Step 1:

Go to https://thingspeak.com/, register an account and login to the platform

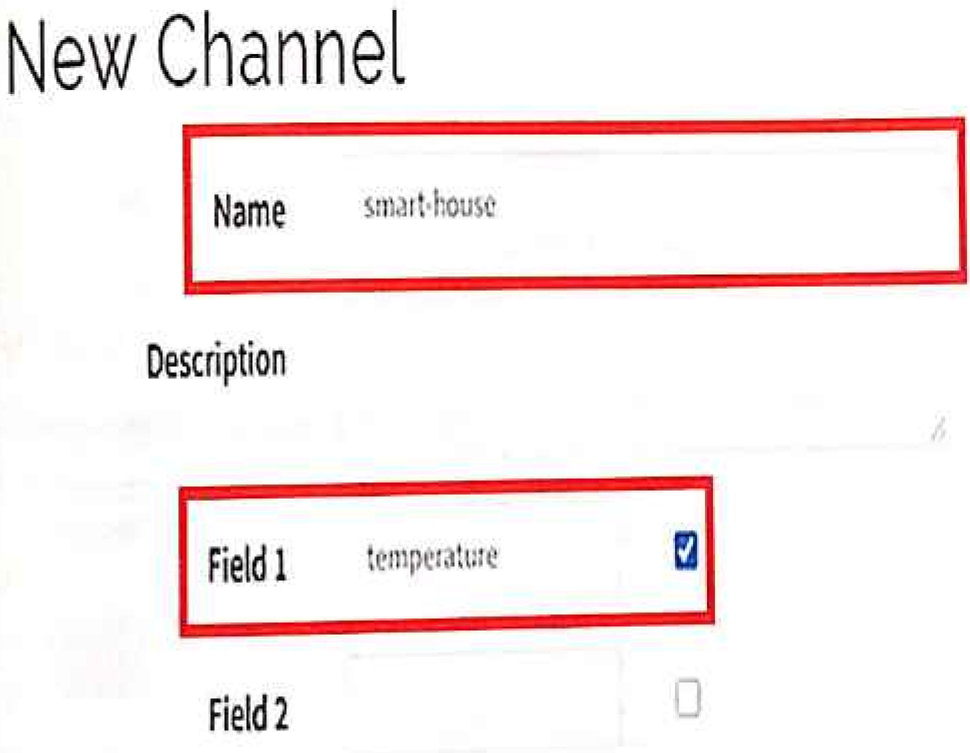
Step 2:

Choose Channels -> My Channels -> New Channel

Step 3:

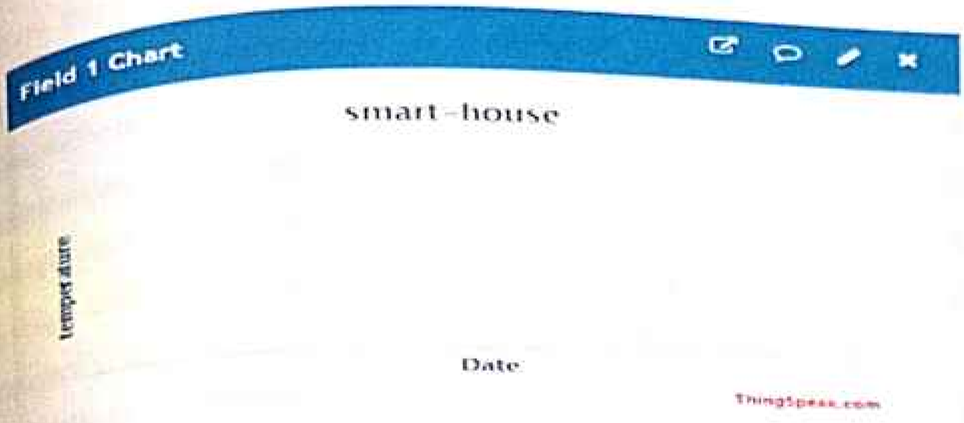
Input Channel name, Field1 , then click Save Channe ll

- Channel name: smart-house
- Field 1: Temperature



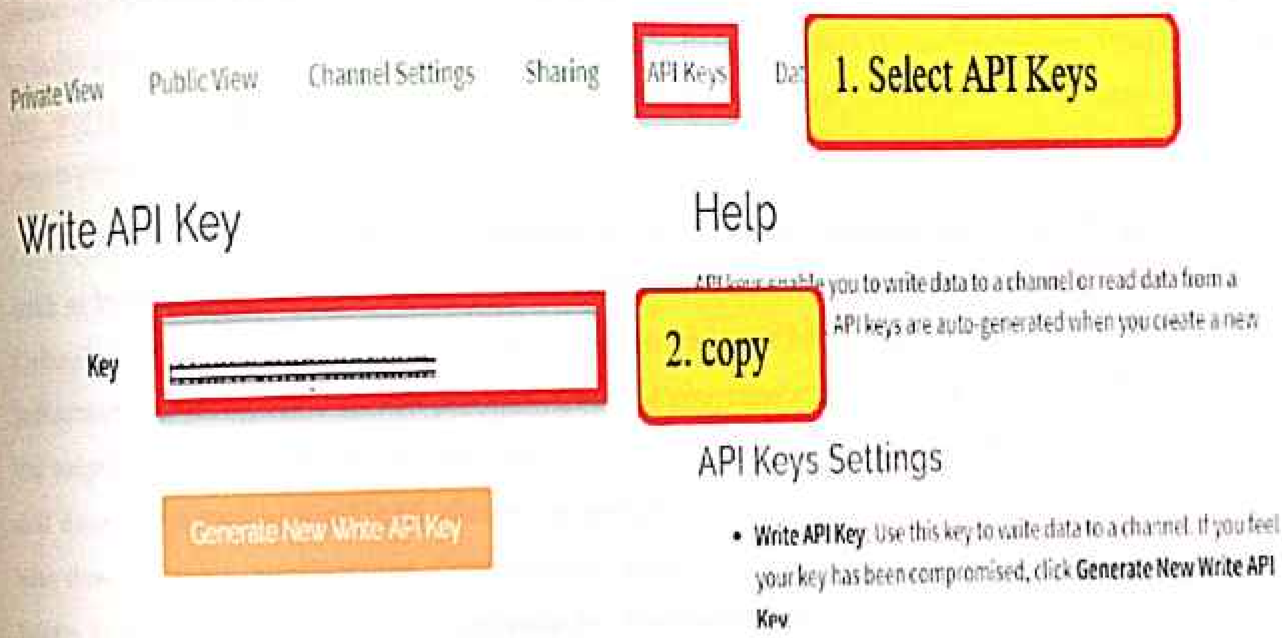
Step 4:

You will see a chat for data field1



Step 5:

Open your web browser, go to https://thingspeak.com , select your channel > –API Keysll ,
copy the API key as follows:



4.4 : MIT Appinventor

The smartphone is an information nexus in today's digital age, with access to a nearly infinite supply of content on the web, coupled with rich sensors and personal data. However, people have difficulty harnessing the full power of these ubiquitous devices for themselves and their communities. Most smartphone users consume technology without being able to produce it, even though local problems can often be solved with mobile devices. How then might they learn to leverage smartphone capabilities to solve real-world, everyday problems? MIT App Inventor is designed to democratize this technology and is used as a tool for learning computational thinkingin a variety of educational contexts, teaching people to build apps to solve problems in their communities. MIT App Inventor is an online development platform that anyone can leverage to solve real-world problems.

It provides a web-based –What you see is what you get‖ (WYSIWYG) editor for building mobile phone applications targeting the Android and iOS operating systems. It uses a block-based programming language built on Google Blockly (Fraser, 2013) and inspired by languages such as StarLogo TNG (Begel & Klopfer, 2007) and Scratch (Resnick et al., 2009; Maloney, Resnick, Rusk, Silverman, & Eastmond, 2010), empowering anyone to build a mobile phone app to meet a need. To date, 6.8 million people in over 190 countries have used App Inventor to build over 24 million apps. We offer the interface in more than a dozen languages. People around the world use App Inventor to provide mobile solutions to real problems in their families, communities, and the world.

The platform has also been adapted to serve requirements of more specific populations, such as building apps for emergency/first responders (Jain et al., 2015) and robotics (Papadakis & Orfanakis, 2016). In this chapter, we describe the goals of MIT App Inventor and how they have influenced our design and development—from the program's inception at Google in 2008, through the migration to MIT, to the present day. We discuss the pedagogical value of MIT App Inventor and its use as a tool to teach and encourage people of all ages to think and actcomputationally. We also describe three applications developed by students in different parts of the world to solve real issues in their communities. We conclude by discussing the limitationsand benefits of tools such as App Inventor and proposing new directions for research.

4.5 : Final Code In Arduino

```
#include <LiquidCrystal.h>
const int rs =8, en =9, d4 =10, d5 =11, d6 =12, d7 =13;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
int m1=3;
int m2=4;
int t,h;
int sm=A0;
int Rs=A3;
int ls=A2;
int led=A1;
int smk=A4;
const int pingPin =6; // Trigger Pin of Ultrasonic Sensor
const int echoPin =5; // Echo Pin of Ultrasonic Sensor
long duration;
int distance;
#include "DHT.h"
#define DHTPIN 7
#define DHTTYPE  DHT11
DHT dht(DHTPIN, DHTTYPE);
int buz=2;
void setup() {
lcd.begin (16,2);
lcd.setCursor(4,0);
lcd.print("WELCOME");
lcd.setCursor(0,1);
lcd.print("SMART AGRICULTURE:");
delay(2000);
lcd.clear();
Serial.begin(115200);
//Serial.begin(9600);
```

```
pinMode(pingPin,OUTPUT);
pinMode(echoPin,INPUT);
dht.begin();
pinMode(smk,INPUT);
pinMode(buz,OUTPUT);
digitalWrite(buz,0);
pinMode(sm,INPUT);
pinMode(ls,INPUT);
pinMode(led,OUTPUT);
pinMode(Rs,INPUT);
pinMode(m1,OUTPUT);
pinMode(m2,OUTPUT);
digitalWrite(m1,0);
digitalWrite(m2,0);
wifi_init();
}
void loop() {
digitalWrite(pingPin, LOW);
delayMicroseconds(2);
digitalWrite(pingPin, HIGH);
delayMicroseconds(10);
digitalWrite(pingPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance= (duration/2)/29.1;
int h = dht.readHumidity();
int t = dht.readTemperature();
int  mval=analogRead(sm);
int rval=analogRead(rs)/10;
int lval=analogRead(ls);
int sval=analogRead(smk);
lcd.clear();
lcd.setCursor(0,0);
```

```
lcd.print("T:"+String(t)+ " H:"+String(h)+ " M:"+String(mval));
lcd.setCursor(0,1);
lcd.print("R:"+String(rval)+ " I:"+String(lval)+ " L:"+String(distance));
upload_iot(t,h,mval,rval,lval,distance,sval);
if(mval>400)
{
 analogWrite(m1,50);
 digitalWrite(m2,0);
}
else
{
  digitalWrite(m2,0);
  digitalWrite(m1,0);
}
if(lval>200)
{
 digitalWrite(led,0);
}
else
{
 digitalWrite(led,1);
}
}
void wifi_init()
 {
  Serial.println("AT+RST");
  delay(4000);
  Serial.println("AT+CWMODE=3");
  delay(4000);
  Serial.print("AT+CWJAP=");
  Serial.write("");
  Serial.print("project1"); // ssid/user name
```

```
Serial.write("");
Serial.write(',');
Serial.write("");
Serial.print("12345678"); //password
Serial.write("");
Serial.println();
delay(1000);
}
void upload_iot(int x, int y,int z,int p,int q,int r,int s) //ldr copied int to - x and gas copied into -
y
{
  String cmd = "AT+CIPSTART=\"TCP\",\"";
  cmd += "184.106.153.149"; // api.thingspeak.com
  cmd += "\",80";
  Serial.println(cmd);
  delay(1500);
 String getStr ="GET /update?api_key=TKEWQ13QQJLKF9L7&field1=";
  getStr += String(x);
  getStr +="&field2=";
  getStr += String(y);
  getStr +="&field3=";
  getStr += String(z);
  getStr +="&field4=";
  getStr += String(p);
  getStr +="&field5=";
  getStr += String(q);
  getStr +="&field6=";
  getStr += String(r);
  getStr +="&field7=";
  getStr += String(s);
  getStr += "\r\n\r\n";
  cmd = "AT+CIPSEND=";
```

```
cmd += String(getStr.length());
Serial.println(cmd);
delay(1500);
Serial.println(getStr);
delay(1500);
}
```

CHAPTER- 5

RESULT

5.1 Results

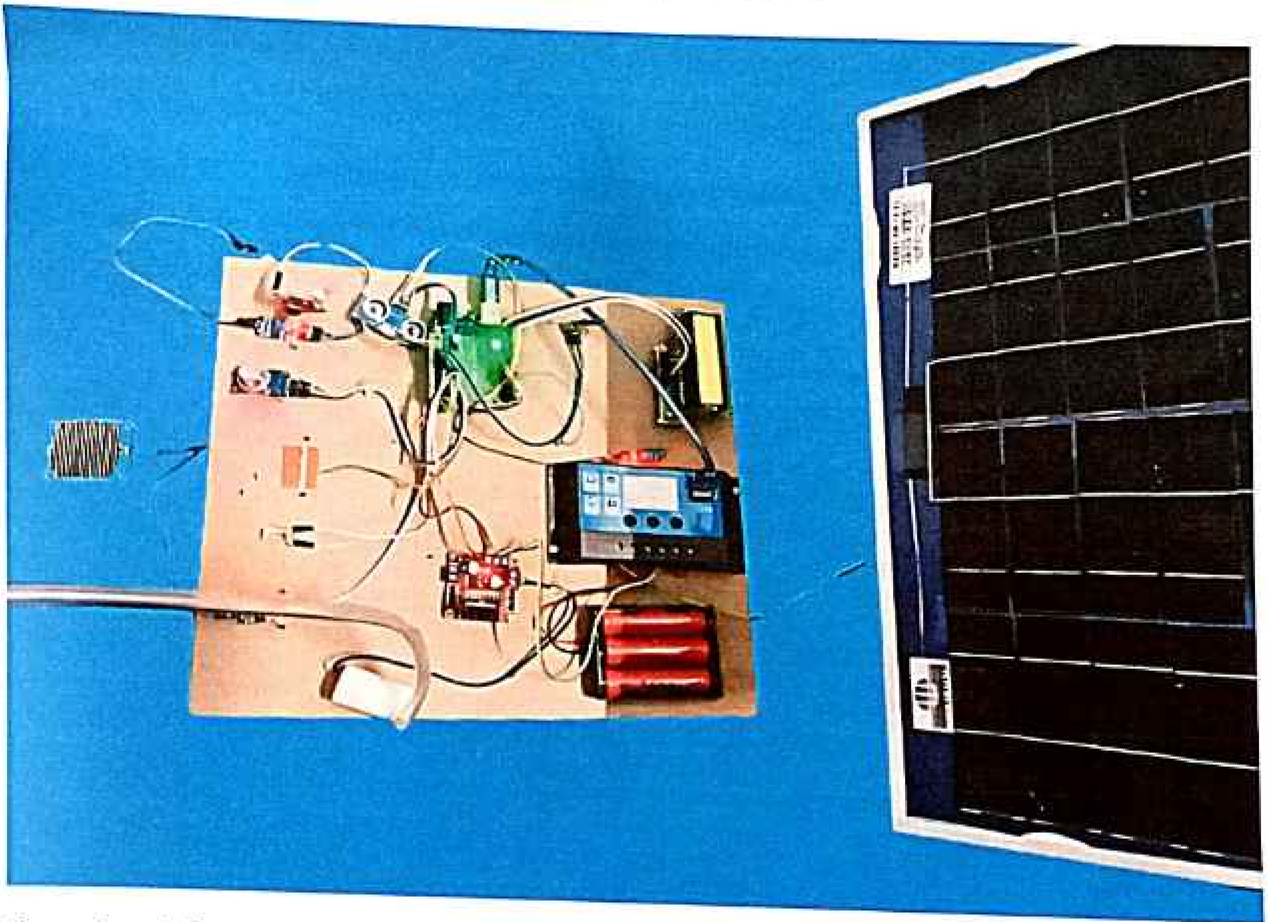
Our IoT agriculture system is meticulously designed to integrate various components seamlessly,ensuring optimal performance and sustainability in agricultural operations. At the heart of the system is an Arduino microcontroller, specifically programmed to oversee data acquisition, actuation, and communication processes. This microcontroller serves as the central processing unit, orchestrating the interactions between sensors, actuators, and communication modules.

For environmental monitoring, the system employs a range of sensors strategically positioned to capture vital parameters essential for crop growth and health. The DHT11 sensor accurately measures temperature and humidity levels, providing crucial insights into ambient conditions within the agricultural environment. Additionally, an analog soil moisture sensor, positioned in the soil, enables precise monitoring of soil moisture content, facilitating targeted irrigationpractices to maintain optimal soil moisture levels for crop growth. Light intensity, a criticalfactor influencing photosynthesis and plant growth, is monitored using an analog light sensor, while the presence of smoke, indicative of potential hazards, is detected through an analog smoke sensor. Furthermore, distance measurement, facilitated by an ultrasonic sensor (HC- SR04), provides valuable information for tasks such as monitoring water levels in reservoirs or detecting the presence of obstacles within the agricultural space.

To translate sensor data into actionable interventions, the system utilizes various actuators. Two DC motors are employed to regulate irrigation systems, adjusting water flow based on real-time soil moisture readings to ensure precise and efficient irrigation. Additionally, an LED light serves as a supplementary lighting source, illuminating when ambient light levels fall below a predefined threshold, thereby providing additional support for plant growth during low-light conditions. These actuators operate in concert with the microcontroller, executing commands promptly to maintain optimal growing conditions for crops throughout the agricultural cycle.

Facilitating seamless communication and data transmission, our IoT agriculture system leverages Wi-Fi connectivity enabled by an ESP8266 module. This connectivity allows for the integration of the system with an IoT platform, such as ThingSpeak, facilitating remote monitoring and analysis of sensor data. Through Wi-Fi, sensor readings are transmitted in real-time to the IoT platform, providing stakeholders with timely insights into environmental conditions and system performance. This remote monitoring capability empowers farmers and agricultural stakeholders to make informed decisions regarding crop management practices, thereby enhancing productivity and efficiency in agricultural operations.

A distinctive feature of our IoT agriculture system is its sustainable and autonomous operation, made possible through the integration of solar power and battery backup. Solar panels harness renewable solar energy to power the system, reducing reliance on conventional energy sources andminimizing the environmental footprint of agricultural operations. Batteries serve as backup powersources, ensuring uninterrupted operation during periods of low sunlight or at night, thereby enhancing system resilience and reliability. Furthermore, a charge controller optimizes battery charging and discharging, prolonging battery lifespan and maximizing energy efficiency



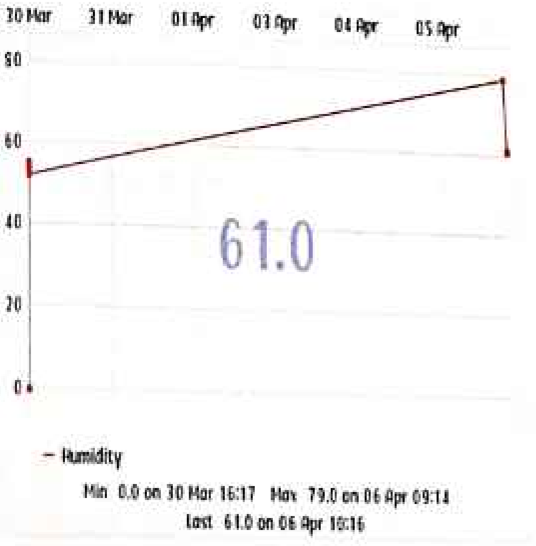
Operational flow within our IoT agriculture system follows a seamless process, beginning with continuous sensor data acquisition to provide a comprehensive overview of environmental conditions. Actuators respond dynamically to sensor readings, implementing targeted interventions to optimize crop cultivation practices in real-time. Data transmission via Wi-Fi enables remote monitoring and analysis, empowering stakeholders with actionable insights to enhance agricultural management practices and optimize resource utilization.

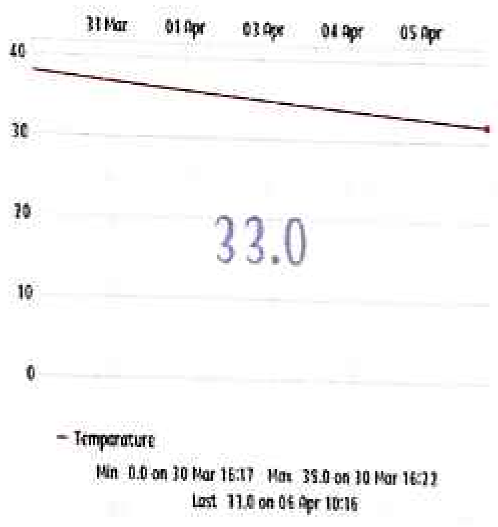
Through rigorous performance evaluation, our IoT agriculture system has demonstrated robust functionality, reliability, and efficiency across diverse agricultural settings. Solar power and battery backup ensure sustained operation, even in challenging environmental conditions, while advanced sensors and actuators enable precise and responsive agricultural management. The integration of remote monitoring capabilities facilitates data-driven decision-making, enhancing productivity and sustainability in agricultural operations.

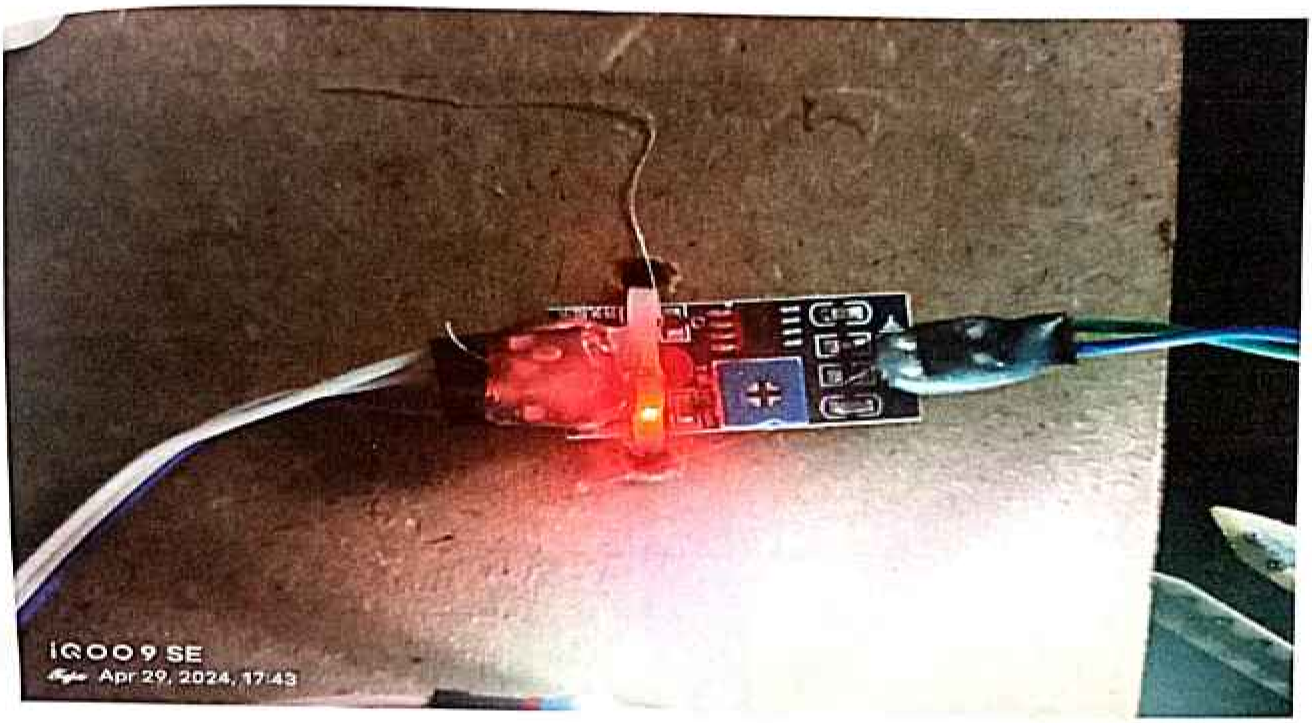
In conclusion, our IoT agriculture system represents a significant advancement in agricultural automation, offering a scalable, sustainable, and technologically advanced solution for modern agriculture. By harnessing renewable energy sources, leveraging advanced sensor technology, and enabling remote monitoring capabilities, the system holds immense potential to revolutionize agricultural practices, promoting efficiency, productivity, and sustainability across agricultural sectors worldwide.

DHT 11 Sensor:

The DHT11 humidity and temperature sensor makes it really easy to add humidity and temperature data. It's perfect for remote weather stations, home environmental control systems, and farm or garden monitoring systems.

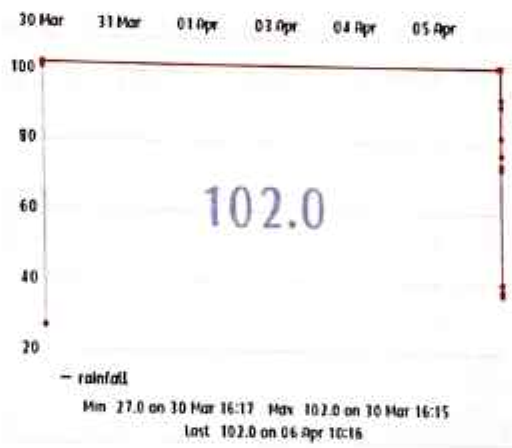






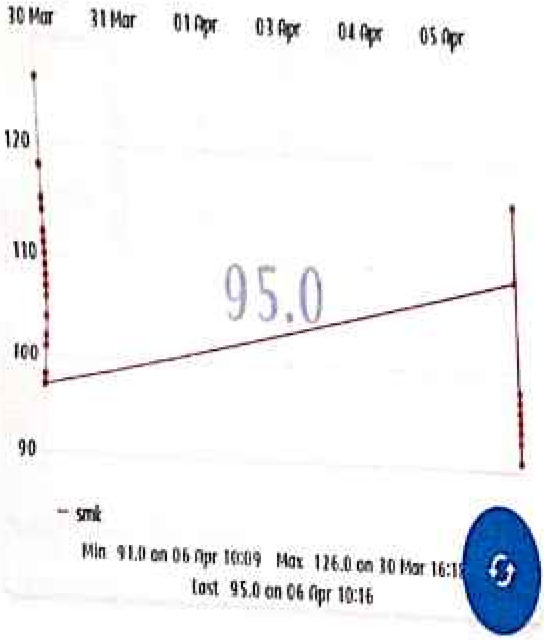
Rain Sensor:

The Arduino Rain Sensor code is very simple and easy to understand. We just have to read the analog data out of the sensor and we can approximate the average rainfall on top of the sensor. This can be easily done in analog mode. In digital mode, we just need to set the potentio meter and we will get a digital output.



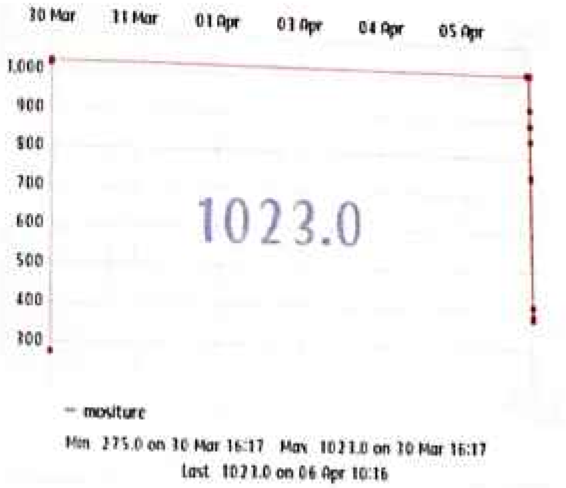
Smoke Sensor:

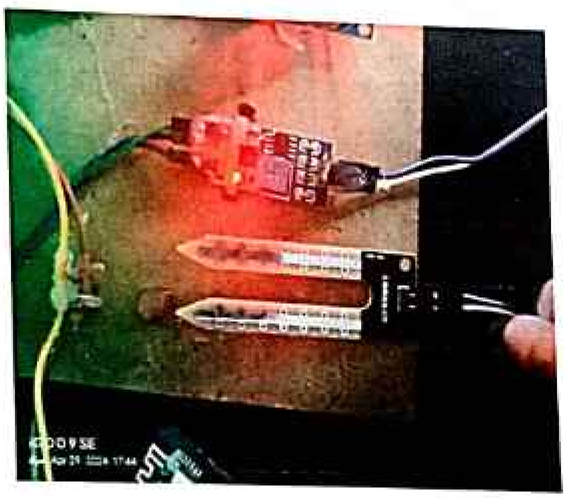
This sensor is also used to detect LPG, i-butane ,propane, methane ,alcohol, hydrogen ,smoke. There by when ever smoke is detected by this sensor it gives information to arduino,and the programme in the Arduino runs and gives the information that smoke detected and buzzer buzzes in caution of smoke.



Soil Moisture Sensor:

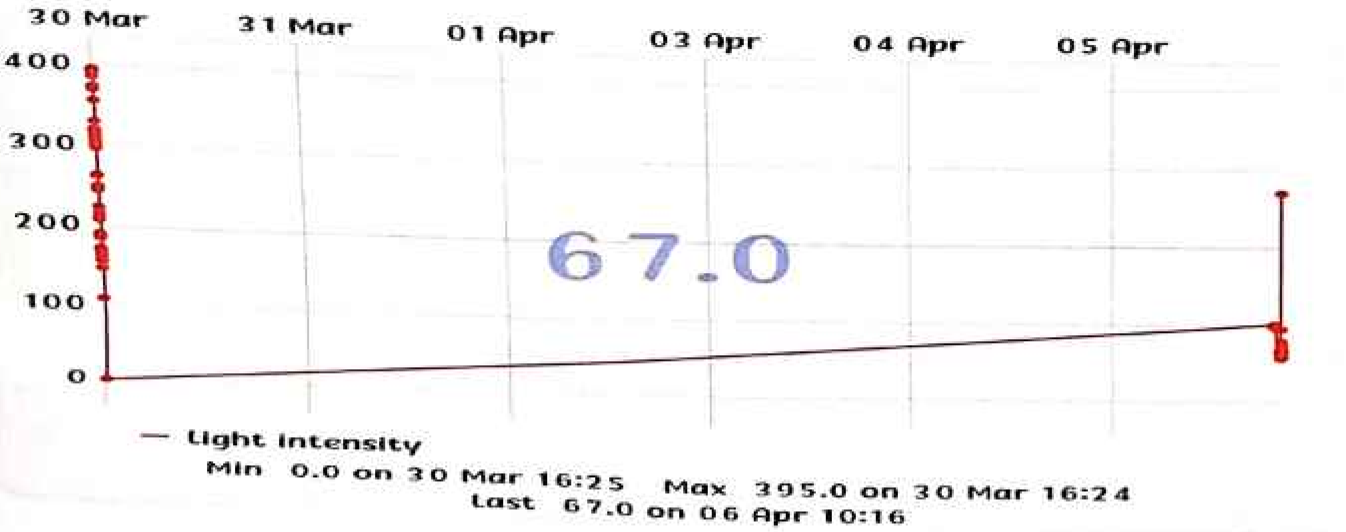
The entire soil humidity sensor consists of two parts: the first one is the soil moisture sensor probe and the second one is an electronic module. The module processes the incoming data from the probe and that gets processed by a microcontroller like Arduino and we get the final output.





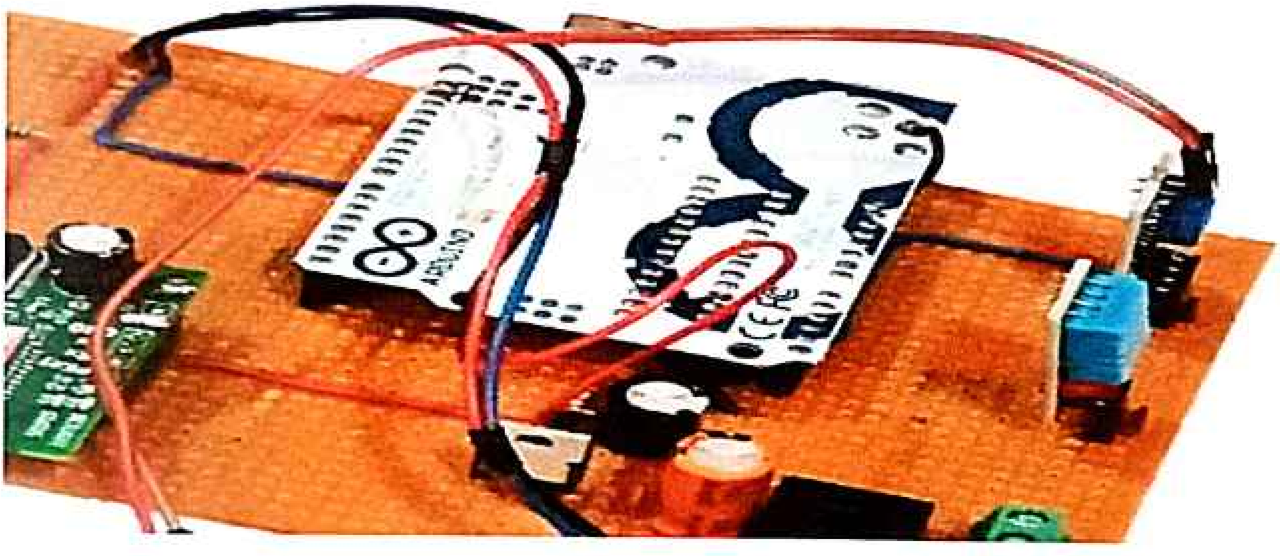
Light Intensity:

The Grove Light Sensor is a module that is used to measure light intensity. It is an analog component, with values ranging between 0 and 1023. The light sensor is a useful component for various reasons, such as detecting movement, tracking the movement of the sun and automatic lighting.



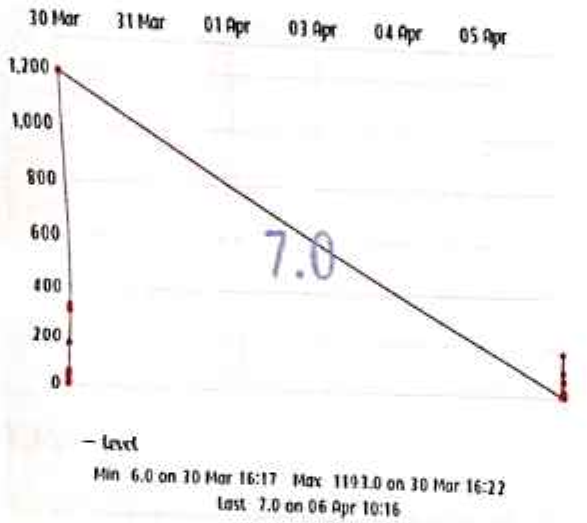
Arduino UNO:

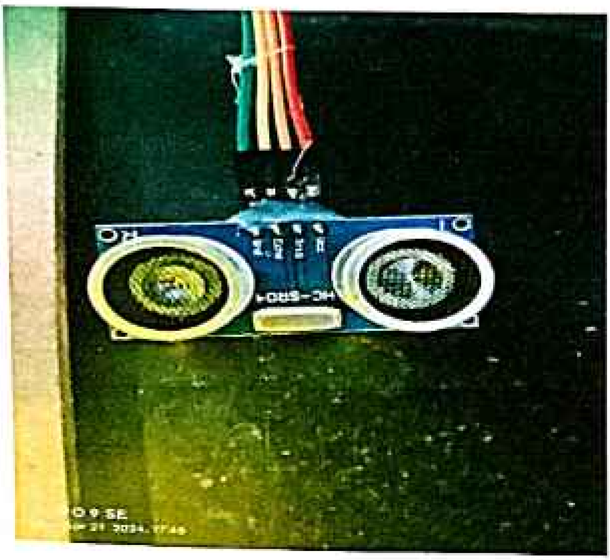
Arduino Uno is open source microcontroller board that helps create interactive projects giving smart solutions by automation. It is based on the processor ATmega328p. It also comes with a variety of input and output pins that can be used to connect different electronic components.



Ultrasonic Sensor:

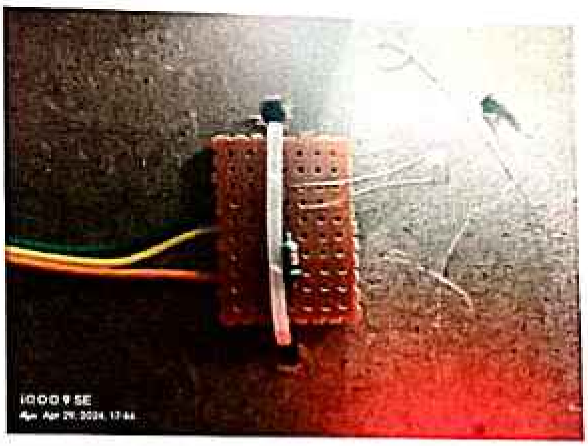
The sensor has 4 pins. VCC and GND go to 5V and GND pins on the Arduino, and the Trig and Echo go to any digital Arduino pin. Using the Trig pin we send the ultrasound wave from the transmitter, and with the Echo pin we listen for the reflected signal.





Light Dependent Resistor:

The LDR Sensor Module is used to detect the presence of light / measuring the intensity of light. The output of the module goes high in the presence of light and it becomes low in the absence of light. The sensitivity of the signal detection can be adjusted using a potentiometer.



Light Emitting Diode:

An LED is a simple diode that emits light in a forward bias. We will write an LED-blinking program on the Arduino IDE and download it to the microcontroller board. The program simply turns ON and OFF LED with some delay between them.

Liquid Crystal Display:

One common type of LCD display used with the Arduino UNO is the 16x2 LCD module, which has 16 columns and 2 rows of characters. To integrate the Arduino UNO with an LCD display, first make sure that each pin is connected on the breadboard correctly, and then place the LCD screen and potentiometer on that breadboard.

CONCLUSION AND FUTURE SCOPE

Conclusion:

It can be concluded that the use of electronic and mechanical System will be very advantageous for better agricultural output. In this project we have designed a model to help the farmers by sending alert messages and controlling agricultural activities in the land in the presence or absence of the farmer using wireless Sensor network technology by simply sending a message. The proposed System is capable of controlling the essential parameters necessary for growth of the plants which includes, monitoring the crops. The microcontroller and sensors are interfaced successfully and wireless communication is achieved. Also, this proposed System of farming is user-friendly, cost effective and highly robust.

From the proposed system we can say that no animal can enter into the farm because of sensors and buzzer are placed if the animal doesn't leave even after the buzzer sound the we can monitor our farm using the camera fixed in the farm and also a notification will be delivered to our mobile so we can go to the farm and make sure that the animal will leave.

Future Scope

The project has vast scope in developing the system and making it more user friendly and the additional features of the system like:

- By installing a webcam in the system, photos of the crops can be captured and the data can be sent to database.
- Speech based option can be implemented in the system for the people who are less literate.
- GPS (Global Positioning System) can be integrated to provide specific location of the farmer and more accurate weather reports of agriculture field and garden.
- Regional language feature can be implemented to make it easy for the farmers who are aware of only their regional language.
- This system claims that irrigation systems become more independent with IOT's rapid data transmission. The primary benefit of IOT is that data is sent even when clients arenot connected to the node network, and whenever a client is connected to that node, they are able to view the data that has already been sent and also can monitor their farm with the camera placed in the farm. So that they can analyse the daily changes in the farm due to animals and increase crop production. Also, this technique will assist them in increasing crop production, which will benefit their economic security.

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